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STATEMENT OF AUTHENTICITY (DECLARATION)

I hereby declare that ***Car Rental System*** is all my own work and that it has not been awarded for any diploma or any other title of any kind. All the sources I have used or quoted have been indicated and acknowledged by complete reference.

Name: Mulenga Chiwele (**author**)

Signature:

Date:

Supervisor's Declaration

I hereby declare that the presentation and preparation of the dissertation were supervised in accordance with the guidelines on the supervision laid down by Evelyn Hone College.

Name: Mrs. Mumba

Signature:

Date:

ABSTRACT

The **Car Rental System** is designed to enable customers to book vehicles from anywhere in the world with ease. This application allows users to enter their details and preferences through a user-friendly online interface. Once registered, customers can quickly reserve the cars they need, tailored to their specific requirements.

In Zambia and many African countries, individuals often rely on outdated methods to find car rentals, spending excessive time searching with limited success. The **proposed system** aims to eliminate these challenges by providing a centralized platform where users can browse a wide selection of vehicles from the comfort of their homes using a computer or smartphone. This automated solution will significantly enhance the efficiency and effectiveness of car rentals in Zambia.

Ultimately, the goal of this system is to provide a convenient online solution for customers to book cars, transforming the traditional rental experience into a seamless and modern process.

DEDICATION

This project is dedicated to my lovely family, whose unwavering support, encouragement, and belief in my abilities have been the cornerstone of my academic journey. Your endless love, sacrifices, and guidance have shaped me into the person I am today, motivating me to pursue my goals relentlessly.

To my friends, I appreciate the camaraderie, inspiration, and countless moments of joy we've shared. Your support during challenging times has been invaluable, and I am grateful for your presence.

Together, you have all contributed to my success, and I hope this work reflects the love and dedication you have shown me.

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Finally, I would like to thank my friends, fellow researchers, and family members for their encouragement and assistance during challenging times. Your support has been instrumental in my research efforts.

Chapter one

Online Car Rental System

1.1 Introduction

In an increasingly mobile world, the demand for efficient transportation solutions is more pressing than ever. Car rental systems have evolved significantly across various regions globally, facilitating seamless travel for individuals and businesses alike. In countries like the United States and the United Kingdom, established platforms such as Enterprise and Hertz leverage advanced technology to provide customers with intuitive booking experiences, real-time vehicle tracking, and tailored services that cater to diverse needs—from leisure trips to corporate events.

In Europe, car-sharing initiatives have gained traction, allowing users to rent vehicles by the hour or day through intuitive mobile applications. This not only promotes convenience but also aligns with growing environmental concerns, as shared vehicles can reduce the number of cars on the road, contributing to decreased traffic congestion and lower emissions.

In Africa, the car rental landscape is also evolving. Countries such as South Africa, Kenya, and Nigeria have witnessed a surge in online car rental platforms that offer not only convenience but also a variety of vehicles to meet different customer demands. These systems typically feature user-friendly interfaces, secure payment gateways, and mobile access, allowing customers to book cars effortlessly. However, despite these advancements, many regions in Africa, including Zambia, still face significant challenges. Traditional car rental services often rely on manual processes, leading to inefficiencies and a lack of real-time information, which can hinder customer satisfaction.

In Zambia, the car rental market is gradually beginning to adapt to these modern solutions. Traditional rental services often depend on manual processes and fragmented booking systems, leading to inconsistencies in vehicle availability and customer service. While some local providers have started to implement online systems, many still operate with limited technological infrastructure, which restricts their ability to offer comprehensive, user-centric services. This

proposal outlines my commitment to developing an advanced Online Car Rental System tailored to address these regional needs, enhancing both user satisfaction and operational efficiency in Zambia's growing market.

1.2 Background

The current car rental industry in Zambia faces significant challenges that hinder its growth and effectiveness, creating a pronounced gap between the ideal user experience and existing services. First, the booking process is often cumbersome and inefficient. Many rental agencies rely on outdated methods, such as phone calls and in-person visits, which can lead to long wait times and user frustration. This inefficiency not only deters potential customers but also results in lost business opportunities, as users increasingly turn to platforms that offer instant online booking.

Second, issues with vehicle availability are common due to the reliance on manual inventory management systems. This can result in inaccuracies, such as double bookings or miscommunication regarding available vehicles. When customers arrive expecting to pick up a car that is not available, it damages the reputation of rental agencies and diminishes customer trust. A lack of real-time updates exacerbates these problems, leaving customers in the dark about their options.

Third, security concerns around online transactions remain a critical issue. Many existing platforms do not implement modern encryption and security protocols, exposing users to risks of fraud and data breaches. This lack of trust in the payment process can discourage potential customers from engaging with rental services online, further limiting the market's growth.

Additionally, the current systems often lack personalization features that can enhance user engagement. Many platforms provide a one-size-fits-all experience, failing to cater to the unique preferences and needs of individual users. This lack of customization diminishes user satisfaction and loyalty, making it difficult for rental companies to retain customers in a competitive market.

Moreover, customer support is often inadequate, with limited channels for assistance. This can lead to unresolved issues and a negative overall experience for users, further contributing to dissatisfaction and loss of business. The reliance on traditional manual methods, such as using Excel sheets for tracking bookings, creates slow processing times and increases the risk of errors. Furthermore, the existing system does not facilitate direct feedback from users to administrators, leading to inconsistencies in service quality.

In summary, the car rental industry in Zambia is characterized by fragmented and outdated practices, which create a substantial gap between the current user experience and the ideal scenario. By addressing these critical challenges—inefficient booking processes, inaccurate vehicle availability, security vulnerabilities, lack of personalization, and inadequate customer support—our proposed Online Car Rental System aims to create a cohesive, secure, and user-friendly platform. This transformation is essential for enhancing customer satisfaction, improving operational efficiency, and positioning Zambian rental services to meet the demands of a rapidly evolving market.

1.3 Objectives

The main objectives of the Online Car Rental System are:

- ❖ To allow users to make secure online payments for rentals.
- ❖ To display real-time car availability and prevent double bookings.
- ❖ To manage detailed vehicle information for operational efficiency.
- ❖ To provide a platform for users to track car rentals.

1.4 Summary

The main objective of this Online Car Rental System project will enable the user to rent a car. The user shall login to the system and check for availability of cars. The user specifies a type of car and the journey date and time. The Car Rental System shall check for the availability of the car and rent the car to the customer. All the data regarding the rental cars are stored in MySQL database. The user has to enter his name, address, phone details and check for the cars available for rent. The UI is very simple and the connectivity to back end is robust. The main advantage is that the user shall be able to choose a car depending on his budget.

1.5 Scope

This project traverses a lot of areas ranging from business concept to computing field, and required to perform several researches to be able to achieve the project objectives.

The area covers include:

Car rental industry: This includes study on how the car rental business is being done, process involved and opportunity that exist for improvement.

General customers as well as the company's staff will be able to use the system effectively.

1.6 Problem Statement: Existing System

The car rental industry is grappling with significant challenges due to outdated systems that hinder customer engagement and operational efficiency. Prospective renters must visit physical offices to register, limiting their ability to view available options in advance and compromising their travel comfort. The existing system lacks direct feedback channels, leading to inconsistencies in service quality.

The reliance on manual processes results in slow processing times and increased risks of double bookings. Additionally, the current system operates only during office hours, restricting customers' opportunities to make transactions or reservations. These issues create a substantial gap between the desired seamless, secure, and personalized car rental experience and the fragmented, inefficient systems currently in place.

Proposed System

The proposed system facilitates the customers to fill up their details, and to give a brief description of a car they want to book. This new system is very helpful for customers who want to hire their cars through this site.

This Car Rental System project will enable the user to rent a car. The user shall login to the system and check for availability of cars. The user specifies a type of car and the journey date and time. The Car Rental System shall check for the availability of the car and rent the car to the customer. All the data regarding the rental cars are stored in MySQL database. The user has to enter his name, address, phone details and check for the cars available for rent. The UI is very simple and the connectivity to back end is robust. The main advantage is that the user shall be able to choose a car depending on his budget.

Advantages:

- ❖ First the customer has to make a reservation and later on in the process has to do registration.
- ❖ Second if the customer had already registered himself then he can continue booking in his own account by giving his customer id or mail id.
- ❖ Thirdly, the customer can amend details or update his details.
- ❖ Maintenance is easy and performance is good
- ❖ It is easy to use and understand.
- ❖ It reduces the time complexity.

Budget

The total estimated budget for the Online Car Rental System is **ZMW 9,350**. The breakdown is as follows:

ITEM	Cost (ZMW)
Personnel Costs	500
Xampp Server Software	Free
Project Management	400
Technology and Equipment	6,000
Data Collection	600
Software Licenses	Free
Visual studio Code	Free
User Training and Documentation	350
Contingency	1,000
Unforeseen Expenses	500
Total Budget	9,350 ZMW

Project Plan

Phases and Timeline

- ❖ The project is planned to start in July 2024 and conclude by the first week of November 2024. Below is a detailed project plan, including durations for each phase:

Phase 1: Planning and Analysis (July 16, 2024 - July 28, 2024)

- ❖ Requirements gathering
- ❖ System design and architecture
- ❖ Initial project setup

Phase 2: Development (July 29, 2024 - September 22, 2024)

- ❖ Frontend development
- ❖ Backend development
- ❖ Integration of PHP and MySQL components

Phase 3: Testing and Quality Assurance (September 23, 2024 - October 10, 2024)

- ❖ System testing
- ❖ Bug fixing
- ❖ Performance optimization

Phase 4: Deployment and Training (October 11, 2024 - October 24, 2024)

- ❖ Deployment to live environment
- ❖ User training
- ❖ System documentation

Phase 5: Maintenance and Support (October 25, 2024 - November 12, 2024)

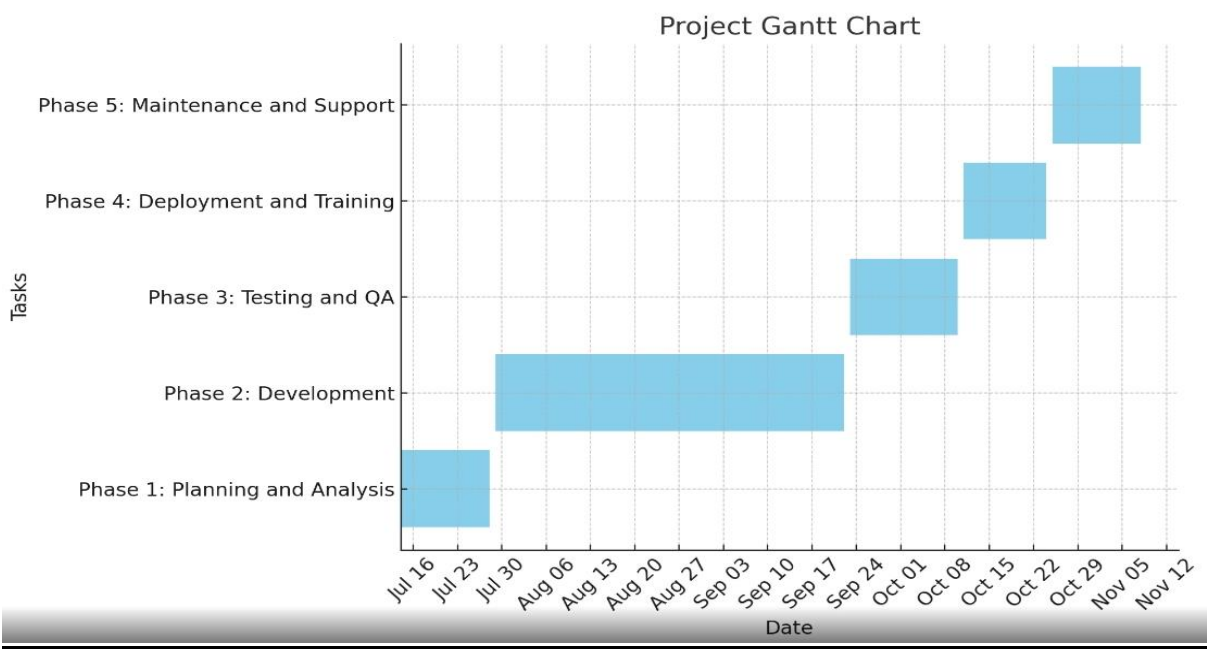
- ❖ Ongoing support
- ❖ Monitoring and maintenance

8.1 Gantt Chart

The Gantt Chart below provides a brief overview of the project timeline:

Task	Start Date	End Date	Duration
Phase 1: Planning and Analysis	July 16, 2024	July 28, 2024	2 weeks
Phase 2: Development	July 29, 2024	September 22, 2024	8 weeks
Phase 3: Testing and QA	September 23, 2024	October 10, 2024	2.5 weeks
Phase 4: Deployment and Training	October 11, 2024	October 24, 2024	2 weeks
Phase 5: Maintenance and Support	October 25, 2024	November 12, 2024	2 weeks

Gantt Chart Diagram



Conclusion

The **Online Car Rental System** will revolutionize the car rental industry by offering a modern and efficient platform for users. With a well-defined project plan and a budget of ZMW 9,350, the system is designed to provide a user-friendly experience and cost-effective solutions. We are confident in the project's success and look forward to delivering a high-quality service that enhances the car rental experience.

Chapter 2

LITERATURE REVIEW

2.0 INTRODUCTION

This chapter presents an in-depth exploration of various car rental systems that closely align with the proposed solution. The analysis focuses on their design, functionality, and effectiveness in meeting objectives, uncovering valuable insights, best practices, and areas for improvement. By contrasting these existing solutions with the new approach, the unique contributions and innovations of the proposed system can be articulated more clearly.

PeeJay's Car Hire operates a user-friendly online booking platform that allows customers to select their preferred vehicles and specify rental dates, showcasing a commitment to customer satisfaction (PeeJay's Car Hire, n.d.). However, the platform lacks a personalized user accounts for saving preferences and rental history, making repeat bookings less convenient. Additionally, the limited payment options restrict flexibility for customers, as it does not accommodate digital wallets or flexible payment plans and that could present challenges for customers seeking a more versatile transaction experience. (PeeJay's Car Hire, n.d.). *Car Rental Services in Zambia*. Retrieved from www.peejayscarhire.com

Benmak Car Hire offers a range of vehicles for short and long-term rentals through an easy online reservation system that emphasizes secure payment options (Benmak Car Hire, n.d.). Despite these strengths, the platform does not fully utilize advanced user interface designs to streamline navigation, which can affect user experience.

The system lacks personalized features, such as user accounts that allow customers to save their preferences and rental history, making repeat bookings less convenient. Additionally, the lack of a customer account management system further limits the overall functionality of the platform. Furthermore, the platform relies primarily on digital marketing channels, such as Facebook, which may restrict its reach to potential customers beyond these platforms. These shortcomings hinder the overall effectiveness of the service and further limit the overall functionality of the system.

(*Benmak Car Hire, n.d.*). Benmak Car Hire. (n.d.). *Reliable Car Rentals for Every Occasion*. Retrieved from www.benmakcarhire.com

Zambia Car Hire is another established player in the Zambian market, providing a variety of vehicles and flexible rental terms. Their platform features straightforward online bookings and competitive pricing. However, user feedback indicates that their website lacks intuitive navigation, making it challenging for customers to search for specific vehicle types efficiently. Additionally, the platform does not offer advanced filtering options, such as by vehicle features or rental duration, which could enhance the search experience.

Furthermore, their customer service response times have been noted as a concern, with delays that could hinder customer satisfaction and trust. These shortcomings limit the overall effectiveness of their service. Zambia Car Hire. (n.d.). *Affordable Vehicle Rentals in Zambia*. Retrieved from www.zambiacarrentals.com

Hertz car hire is a well-known international car rental company with an easy-to-use online booking system. However, it lacks several important features. First, it doesn't offer personalized accounts, making it hard for customers to repeat their bookings. Second, its payment options lack modern digital methods, which could turn away tech-savvy users. The website also lacks good navigation and filtering options for finding specific types of vehicles, leading to frustration. Finally, Hertz primarily uses traditional marketing, which limits its reach. Hertz. (n.d.). *Rent a Car: Deals on Car Rentals*. Retrieved from www.hertz.com

Comparative analysis with international players such as Bidvest Car Rental and Sixt Rent a Car reveals additional strengths and weaknesses. While these companies provide extensive vehicle choices and robust booking functionalities, their pricing structures may not be competitive for local travelers. Furthermore, their complex booking processes can overwhelm users seeking straightforward rental solutions.

The proposed system aims to address these shortcomings by prioritizing an enhanced user experience through an intuitive design that simplifies the booking process. A customer account management system will be implemented, allowing users to create accounts to save preferences,

view rental history, and manage reservations easily. Furthermore, a focus on a smooth reservation and payment system will be implemented, featuring a variety of payment methods, including credit cards, digital wallets, and other popular options. This flexibility will ensure that transactions are both seamless and convenient.

To increase visibility and attract a broader customer base, targeted digital marketing campaigns will be developed. Utilizing social media platforms for direct engagement with potential customers will effectively showcase unique offerings and build brand awareness, driving traffic to the platform and fostering long-term customer loyalty. By implementing these improvements, the proposed system will not only meet the needs of users but also distinguish itself within the competitive landscape.

Chapter 3

REQUIREMENTS SPECIFICATION / SUCCESS CRITERIA

3.1 INTRODUCTION

This chapter outlines the functional and non-functional requirements for the online car rental system, ensuring alignment with the project's objectives. By specifying these requirements, the aim is to create a user-friendly platform that effectively meets the needs of both customers and car rental operators. The requirements are informed by comprehensive fact-finding methods, including interviews and observations.

3.2 Fact-Finding Methods

Interviews

To gather detailed insights for the online car rental system, interviews were conducted with key stakeholders, including potential users, car rental operators, and industry experts. The primary objective was to understand user expectations and identify pain points in current systems. A semi-structured interview format was employed, allowing for guided questions alongside open-ended discussions. This approach facilitated a deeper exploration of user needs and preferences. Key topics addressed during the interviews included desired features for an online booking system, challenges experienced with existing rental processes, preferences for payment methods and vehicle selection, and expectations for customer support and service reliability. The feedback obtained from these interviews proved invaluable in shaping the functional and non-functional requirements of the system.

Observations

In addition to interviews, observational research was conducted at two local car rental agencies. This method provided firsthand insights into operational processes and customer interactions occurring in real time. Observations focused on several aspects, including the workflow of car rental transactions, customer behavior during the rental process, and interactions with current

booking systems. This method revealed practical challenges faced by both customers and staff, such as inefficiencies in the booking process and common customer queries. The insights gained from these observations complemented the information collected during interviews, offering a comprehensive understanding of the existing landscape and informing the design of the new online car rental platform.

Functional Requirements

Requirement analysis is a crucial software engineering technique that encompasses various tasks aimed at determining the needs and conditions that must be met for a new or modified product, while considering potential conflicts among different users' requirements. Functional requirements describe the internal workings of the system, detailing its functions and processes, the data it should manage, and its user interfaces. The identified functional requirements for the online car rental system include several key components. The system will allow new users to register online and generate a membership card, facilitating easy access and usage. Additionally, customers will be able to make online reservations for cars, streamlining the booking process. Furthermore, the system will automatically update the database whenever a new reservation is made or a new customer registers, ensuring that the data remains current without requiring additional efforts from the administrator. These functionalities are essential for creating an efficient and user-friendly platform.

Non-Functional Requirements

Non-functional requirements describe the aspects of the system that pertain to how well it delivers on the functional requirements. Key non-functional requirements for the online car rental system include security, availability, and ease of use.

First, security is paramount; the system must provide a high level of protection and integrity for the data it holds. Access to sensitive areas of the system should be restricted to authorized personnel only, and users must log in with valid usernames and passwords to view their accounts. Second, the system must ensure high availability, operating 24 hours a day, 7 days a week. In the event of a significant malfunction, the system should be restored within 1 to 2 working days to minimize disruptions to business operations.

Finally, ease of use is crucial, given the diverse levels of technical knowledge among users. A simple yet high-quality user interface should be developed to ensure that users can navigate the system easily, requiring minimal training. Together, these non-functional requirements help ensure that the online car rental platform is reliable, secure, and user-friendly.

Hardware and Software Requirement

Hardware Requirements:

Processor: Intel Core i5-7200U (dual-core, 2.50 GHz or equivalent)

- ❖ Ensures smooth application performance and efficient multitasking.
- ❖ RAM: 8 GB

Supports running multiple applications simultaneously.

- ❖ Hard Disk: 500 GB Space

Provides ample storage for software, projects, and data files.

Software Requirements:

Operating System: Windows /iOS/Unix 17

- ❖ Compatible with various operating systems for flexibility.

Web Browser: Google Chrome/Firefox

- ❖ Required for accessing web applications and ensuring compatibility.

Technology: PHP

- ❖ Required for accessing web applications and ensuring compatibility.

Tools: XAMPP

- ❖ Local server environment for PHP and MySQL development.

Web Design: HTML, CSS, JAVASCRIPT

- ❖ Core technologies for creating and styling web pages.

Back End: MYSQL

- ❖ Server-side scripting language for dynamic content and database interaction.

Scripting Language: PHP

- ❖ Essential for server-side logic in web applications.
- ❖

Chapter 4

DESIGN SPECIFICATION

4.1 INTRODUCTION

The main purpose is to give the architectural description of the system as well as a guide during software development. It gives specification description of how the proposed system will be developed. It describes the operation of the system in terms of modules and how they work together to achieve its functionality, data input, processing as well as outputs.

4.2 DATABASE DESIGN

The design shows how records are going to be stored in the database. The database that has been used for this project is MSQl database. The following tables state the data types that have been used for each attribute. This stage of the database design involves the creation of the actual database structures using the chosen Database Management System (DBMS). In this case, phpAdmin was used.

The tables below show the design of the actual tables in the database:

Table: Admin

Name	Data Type	Size
<u>Id pk</u>	int	11
Username	Varchar	100
Password	Varchar	100
UpdationDate	Timestamp	

The admin table stores admin user information, including a unique identifier (Id, int), username (Username, Varchar), hashed password (Password, Varchar), and a timestamp for the last update (UpdationDate).

Table: User

Name	Data type	Size
<u>Id pk</u>	int	120
Full name	varchar	100
Email id fk	varchar	100
Password	varchar	11
Contact no	char	100
Dod	varchar	255
Address	varchar	100
City	varchar	100
Country	timestamp	
Regdate	timestamp	
Updationdate		

The User table holds essential user information, including a unique identifier (Id), full name, email, hashed password, contact number, date of birth, address, city, and country. It also records registration and last update dates.

Table: Booking

Name	Data Type	Size
<u>Id pk</u>	Int	11
UserEmail	Varchar	100
VechicleId	Int	11
From Date	Date	
ToDate	Date	
Status	Int	11
PostingDate	timestamp	

The Booking table tracks vehicle bookings with fields for a unique ID, user email, vehicle ID, booking dates, status, and posting date.

Table: pages

Name	Data Type	Size
<u>Id pk</u>	Int	11
PageName	Varchar	255
Type	Varchar	255
Detail	longtext	

The Pages table contains page information, including a unique ID, page name, type, and detailed content.

Table: Contact us information

Name	Data Type	Size
<u>Id pk</u>	Int	11
Address	Tinytext	
EmailId	Varchar	255
ContactNo	Char	15

The Contact Us Information table holds details for user inquiries, including a unique ID, address, email, and contact number.

Table: Contact us Query

Name	Data Type	Size
<u>Id pk</u>	int	11
Name	Varchar	100
EmailId	Varchar	120
ContactNumber	Char	11
Message	Longtext	
PostingDate	Timestamp	

Status	Int	11
--------	-----	----

The Contact Us Query table captures user inquiries, including a unique ID, name, email, contact number, message, posting date, and status.

Table: Brands

Name	Data Type	Size
<u>Id pk</u>	int	11
Brand Name	varchar	120
CreationDate	timestamp	
UpdationDate	timestamp	

The Brands table stores brand information, including a unique ID, brand name, creation date, and last update date.

Table: Testimonial

Name	Data Type	Size
<u>Id pk</u>	Int	11
UserEmail	Varchar	100
Testimonial	Mediumtext	
PostingDate	Timeststamp	
Status	Int	11

The Testimonial table records user testimonials, including a unique ID, user email, testimonial content, posting date, and status.

Table: Vehicles

Name	data type	Size
<u>Id pk</u>	int	11
Vehicles Title	varchar	50
Vehicle Overview	longtext	
Price PerDay	float	
Fuel Type	varchar	100
Model Year	int	6
Seating Capacity	int	11
Vimage1	varchar	120
Vimage2	varchar	120
Vimage3	varchar	120
Vimage4	varchar	120
Vimage5	varchar	120
AirConditioner	int	11
PowerDoorLocks	int	11
AntiLockBrakingSystem	int	11
BrakeAssist	int	11
PowerSteering	int	11
DriverAirbag	int	11
PassengerAirbag	int	11
PowerWindows	int	11
CDPlayer	int	11
CentralLocking	int	11
CrashSensor	int	11
LeatherSeats	int	11
RegDate	timestamp	
UpdationDate	timestamp	
VehiclesBrand	int	11

The Vehicles table holds vehicle details, including a unique ID, title, overview, price per day, fuel type, model year, seating capacity, multiple images, features, and safety options. It also tracks registration and update dates, along with the vehicle brand.

Table: Subscribers

Name	Data Type	Size
<u>Id pk</u>	Int	11
SubscrbeEmail	Varchar	120
PostingDate	Timestamp	

The Subscribers table stores information about users who subscribe, including a unique ID, subscriber email, and posting date.

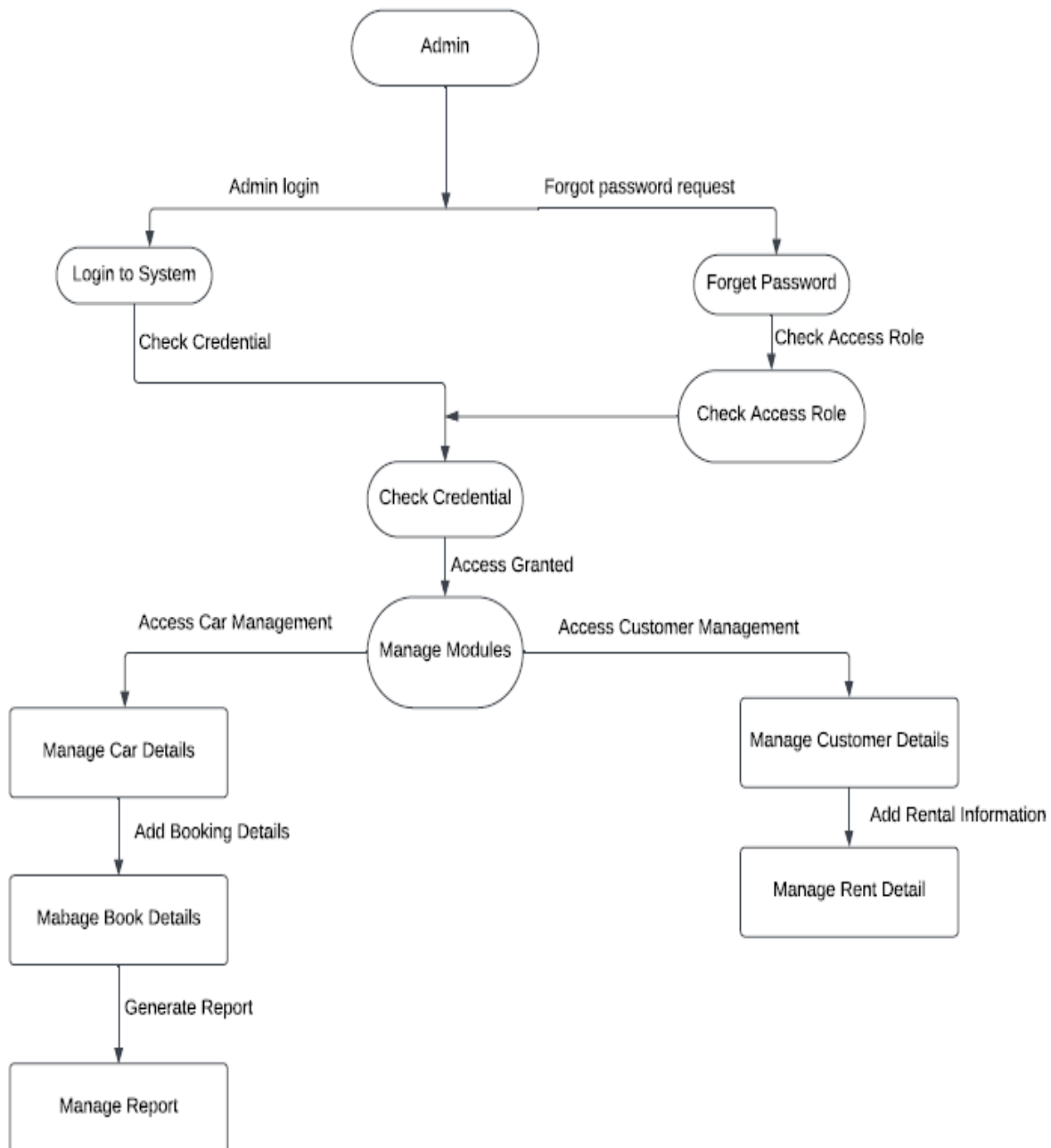
Table: Payment

Name	Data type	Size
<u>Id pk</u>	int	11
Amount	float	
User id	int	11
Vehicle id fk	int	11
Payment date	date	
Booking id fk	int	11

The **Payment** table stores details about payments made by users for vehicle bookings. Each record in this table represents a payment transaction linked to a specific booking and vehicle

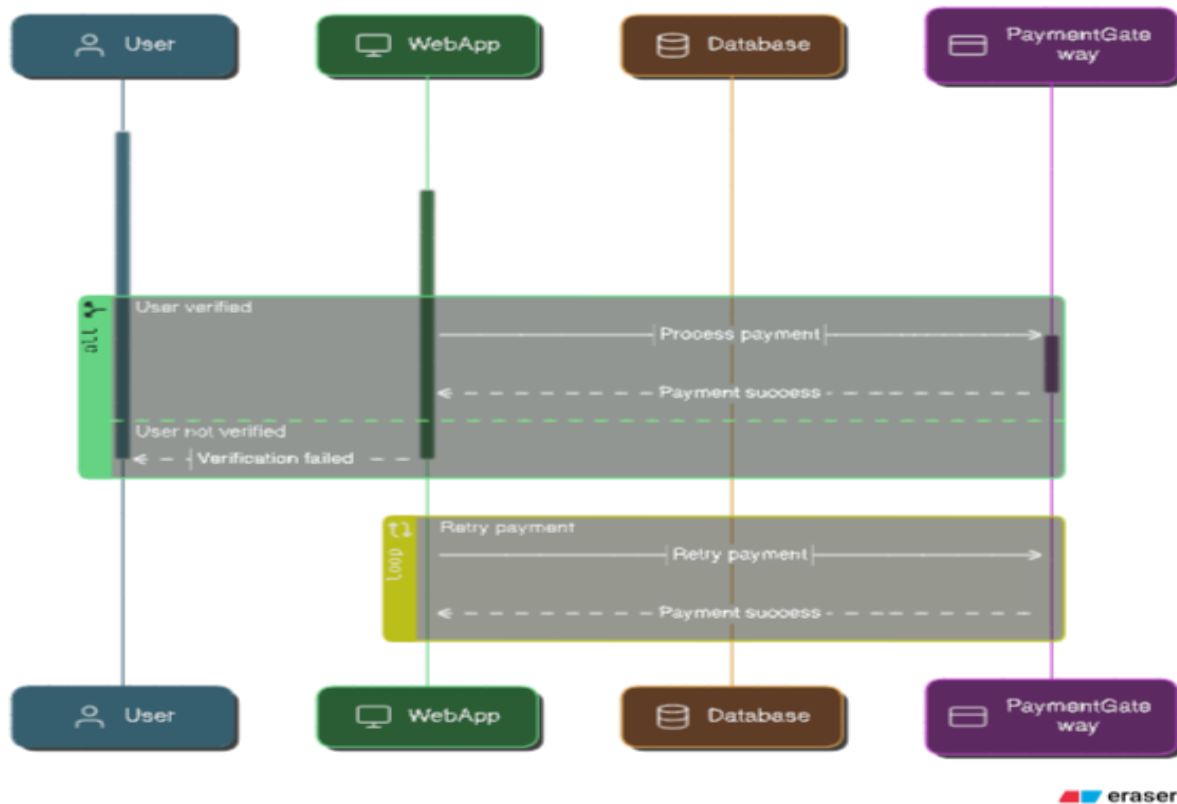
4.3 Context Diagram

The Context Diagram for an Online Car Rental System provides a top-level view of the system, showing the primary interactions between the system and its external entities. This diagram outlines the boundaries of the system, illustrating how data flows between the Online Car Rental System and other connected entities without detailing internal processes.



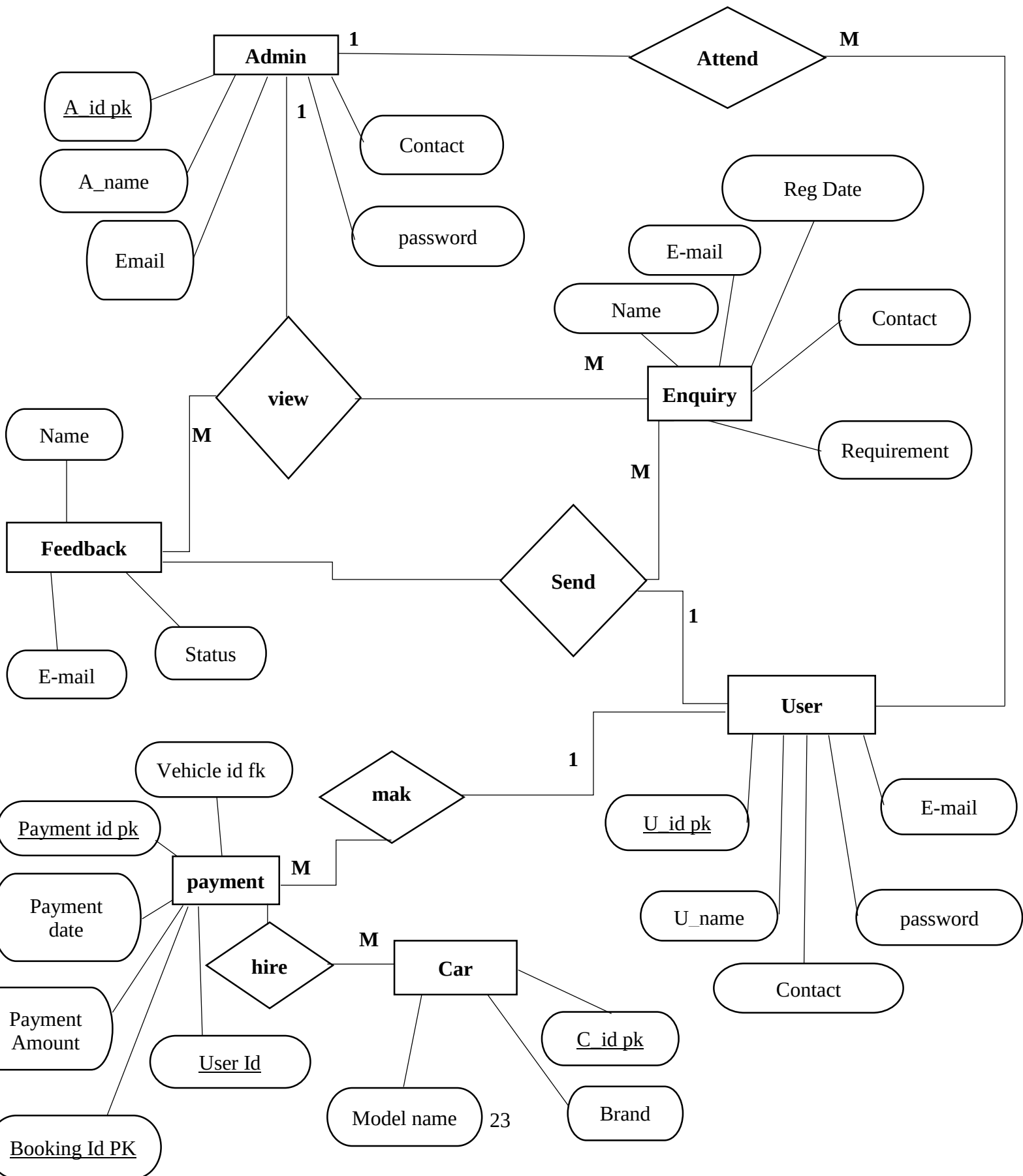
4.5 Sequential Diagram

A Sequential Diagram is a graphical representation that depicts the information flow and the transformations that occur as data moves sequentially through the system, from input to output.



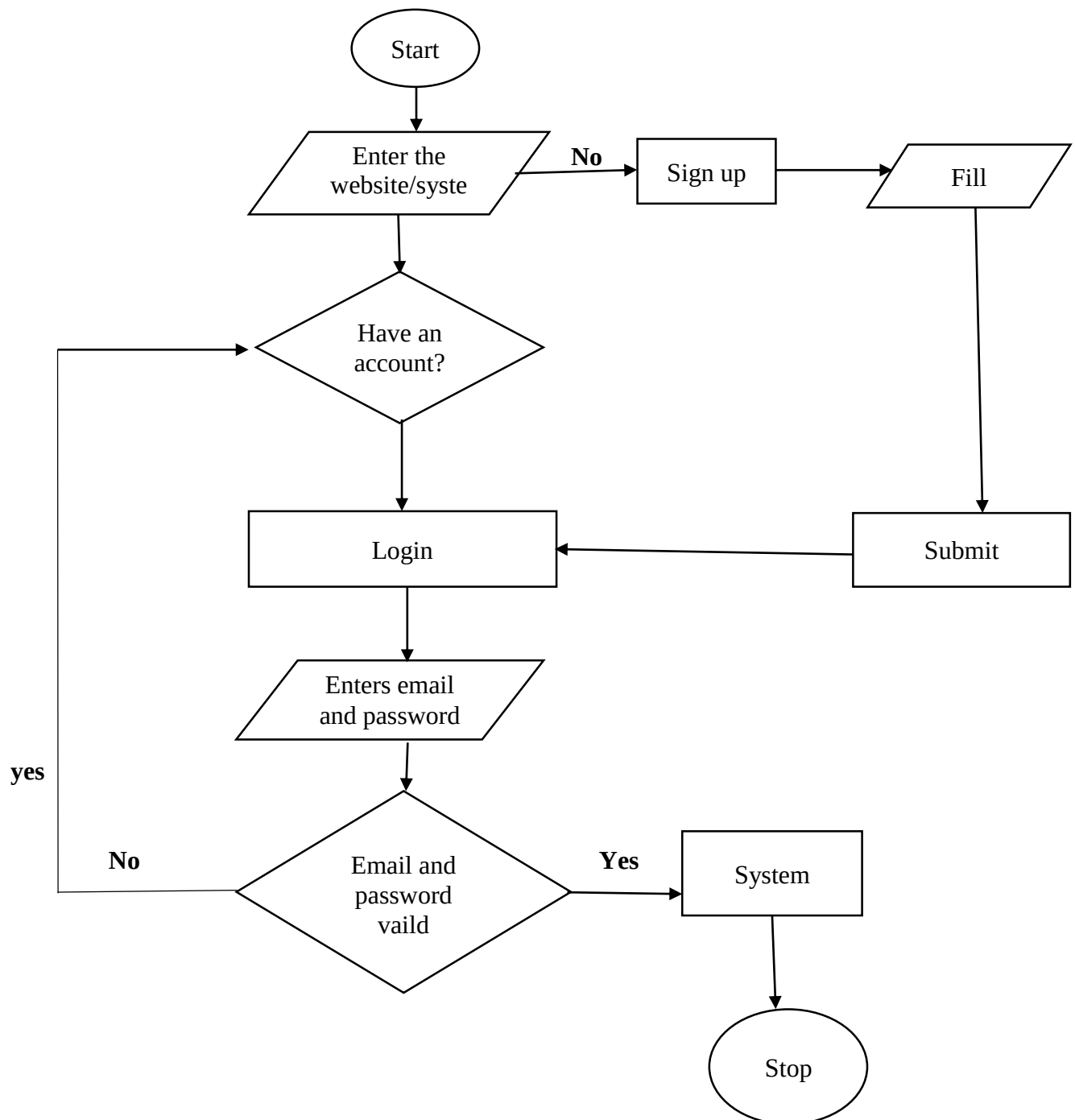
The **Sequential Diagram** provides a high-level overview of the **Online Car Rental System**. It illustrates the major processes and interactions between different components, representing the overall structure of the system. This diagram serves as an overview of the entire online car rental process, highlighting key entities involved in the car rental system.

ERD DIAGRAM



SECURITY DESIGN

Security is an important measure in the implementation of any system as it ensures only registered or authorized users to access the system. Below is a flow illustrating the security flow chart of the system.



USER INTERFACE DESIGN

User interface is the front-end application view to which user interacts with in order to use the software.

The software becomes more popular if its user interface is;

- ❖ Attractive
- ❖ Simple to use
- ❖ Responsive in short times
- ❖ Clear to understand
- ❖ Consistent on all interface screen

The goal of this phase is to define the set of interface objects and actions, referred to as control mechanisms, that allow the user to perform desired tasks. These control mechanisms are crucial in enabling user interaction with the system. It is important to specify how these control mechanisms affect the system's behavior and user experience. Additionally, the phase involves detailing the actions, sequences of tasks, and subtasks, commonly referred to as a user scenario. This includes indicating the system's state when the user performs a particular task, ensuring that the system responds correctly and intuitively at each step.

OUTPUT DESIGN

Input design outlines how users interact with the system through conventional input devices such as keyboards, mice, and printers. A well-designed input system considers various factors, including user needs and preferences, to ensure smooth and efficient user-computer interaction. Proper input design enhances usability, minimizes errors, and improves the overall user experience.

OUTPUT DESIGN

A quality output is one that meets the requirements of the end user and presents information clearly. In any system, the results of processing are communicated to users and other systems through output. Output design determines how information is displayed to address immediate

needs. It is the most vital and direct source of information for users, ensuring that the data is presented in a clear, understandable, and actionable format.

Test Case Table

The following table provides an overview of the test cases conducted to evaluate the main functionalities of the "Car Rentals" system. Each test case describes a specific feature of the system, the input data used, the expected outcome, and the actual outcome after testing. This helps ensure that every critical function within the system meets its design specifications and performs as intended.

Test case ID	Test Description	Input Data	Expected Result	Actual Result	Status
TC- 01	Login with Vaild Details	Vaild Username/Password	User Successfully Logs in	User successfully logs in	Pass
TC- 02	Login with invalid details	Incorrect password	Error message displayed	Error message displayed	Pass
TC- 03	Search for a vehicle	Vehicle model	Relevant search results displayed	Relevant results displayed	Pass
TC- 04	View vehicle details	Vehicle ID	Display detailed vehicle information	Vehicle details displayed	Pass
TC- 05	Make a booking	Vehicle ID, Date	Display detailed vehicle information	Booking confirmation displayed	Pass
TC- 06	Process payment	Credit card, mobile wallet	Payment processed successfully	Payment successful	Pass

This table helps illustrate the thorough testing approach used to validate each major component of the "Car Rentals" system, providing insights into whether each function operates as expected.

Chapter 6

6.0 INTRODUCTION

This chapter provides a detailed overview of the implementation process for the Online Car Rental System, focusing on its development, deployment, and integration into the existing environment. It highlights the technologies, tools, and methodologies used to ensure the system's functionality, user-friendliness, and operational success. Additionally, the chapter discusses the transition from the old manual system to the new computerized one. It also suggests alternative approaches to improve performance, scalability, and effectiveness for future projects.

6.1 Implementation

The project was implemented using a combination of robust technologies and strategies. **PHP 7.4** was chosen for server-side scripting due to its performance, scalability, and security features. For front-end development, **HTML, CSS, and JavaScript** were used to build a responsive, interactive, and user-friendly interface. **Bootstrap** was selected to quickly create a mobile-friendly design with pre-built components, enhancing development speed and ensuring a professional look. The database was managed using **MySQL 7.6**, known for its speed, reliability, and ease of integration with PHP.

A **parallel changeover** method was used for system implementation, allowing both the old and new systems to run simultaneously. This approach minimized risks by allowing users to compare outputs from both systems before fully transitioning to the new one.

The system was installed and quickly set up, followed by thorough training for users, supported by detailed documentation. This ensured that users understood how to operate the system effectively, making the transition smoother.

Overall, the selected tools and methods were chosen to ensure optimal system performance, ease of use, and smooth integration.

The following activities would take place to convert from old manual to new computerized system:

Hardware Requirements:

Processor: Intel Core i5-7200U (dual-core, 2.50 GHz or equivalent)

- ❖ Ensures smooth application performance and efficient multitasking.

RAM: 8 GB

- ❖ Supports running multiple applications simultaneously.

Hard Disk: 500 GB Space

- ❖ Provides ample storage for software, projects, and data files.

Software Requirements:

Operating System: Windows /iOS/Unix 17

- ❖ Compatible with various operating systems for flexibility.

Web Browser: Google Chrome/Firefox

- ❖ Required for accessing web applications and ensuring compatibility.

Technology: PHP

- ❖ Required for accessing web applications and ensuring compatibility.

Tools: XAMPP

- ❖ Local server environment for PHP and MySQL development.

Web Design: HTML, CSS, JAVASCRIPT

- ❖ Core technologies for creating and styling web pages.

Back End: MYSQL

- ❖ Server-side scripting language for dynamic content and database interaction.

Scripting Language: PHP

- ❖ Essential for server-side logic in web applications.

INTERFACE IMPLEMENTATION

The system application software features a clear and user-friendly interface, designed to enhance usability. It consists of independent forms, each performing specific tasks, which allows users to navigate the application with ease. The implementation process utilized PHP and JavaScript as the primary programming languages, ensuring robust functionality and interactivity.

TRAINING PERSONNEL

Training for personnel involved in the system's implementation is essential for understanding their roles, how to effectively use the system, and its capabilities and limitations. This training will equip staff with the necessary knowledge and skills, ensuring a successful transition and maximizing the system's benefits.

SYSTEM CONVERSION

The system will be converted using a direct cutover method. Given that the old system is entirely manual, this approach allows users to quickly adapt to the new system without reliance on previous methods. The cumbersome files of the old system will be replaced with terminals for various operations, improving processes and enhancing efficiency.

FEEDBACK

The feedback received from various users has been overwhelmingly positive. Many expressed that the new system will significantly simplify the process of finding a car rental, allowing them to do so from the comfort of their homes and at their convenience. This ease of access minimizes the effort required to search for rental options, making the experience more enjoyable.

Chapter 7

CONCLUSION

The Online Car Rental System has successfully met all its key objectives, enhancing both user experience and operational efficiency. The system allows users to securely make online payments for rentals, providing a safe and seamless transaction process that ensures peace of mind for customers. Additionally, the system displays real-time car availability, effectively preventing double bookings and improving scheduling accuracy for the rental company. This feature contributes to better resource management and customer satisfaction.

The system also manages detailed vehicle information, ensuring that users have access to comprehensive data on each car. This feature not only helps users make informed rental choices but also supports the company's operational efficiency by maintaining accurate and up-to-date vehicle details. Moreover, the system offers a platform for users to track their car rentals, ensuring transparency and convenience throughout the rental period.

While the system has achieved its objectives, it faced several challenges during development. The integration of secure payment gateways was complex and required careful attention to security protocols. Similarly, the implementation of real-time synchronization for car availability presented technical hurdles, ensuring that information is always accurate and up-to-date. A limitation of the system is that users without internet access remain dependent on physical locations for rental services, limiting the system's accessibility.

In conclusion, the Online Car Rental System has successfully fulfilled its objectives of facilitating secure online payments, real-time car availability tracking, and managing detailed vehicle information. While challenges in system integration and internet dependency remain, the overall impact of the system on operational efficiency and user convenience is significant. Future improvements should focus on overcoming these limitations to further enhance the user experience and expand accessibility.

RECOMMENDATIONS

AS indicated in the problem statement highlighting the inefficiencies faced by the existing systems, I would recommend the adoption and usage of the proposed system as it brings about many benefits to the end user.

However, the system has some limitations that can be improved in the future. Therefore, the following are suggestions that can be done to improve the ability of the system.

Integrating Google Maps will enhance the user experience by providing real-time directions and turn-by-turn navigation within the app. Users will receive live traffic updates to avoid congested routes, while interactive map features will showcase nearby attractions and points of interest, encouraging exploration.

Incorporating GPS functionality will allow users to easily find nearby rental locations by accurately detecting their current positions. Geolocation services will enable dynamic filtering based on vehicle availability and promotions, while smart vehicle suggestions will recommend suitable options tailored to the user's location and destination.

Implementing location-based services will offer augmented reality (AR) navigation assistance for an immersive travel experience. Users will receive promotional notifications for special offers near rental locations and event-based promotions linked to local activities, enhancing engagement and encouraging exploration during their rental period.

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APPENDICES

APPENDIX A

INSTALLATION MANUAL

The installation manual is a guide that shows the steps on how to install the program or system on the user's platform. It also indicates the minimum software and hardware requirements that are needed for the system to be able to run smoothly and efficiently.

1.1 SYSTEM REQUIREMENT

These are the minimum requirements that the user need to have in order for them to be able to run the program efficiently on their platform.

Hardware Requirements:

- ❖ Processor: Intel Core i5-7200U (dual-core, 2.50 GHz or equivalent)
- ❖ RAM: 8 GB
- ❖ Hard Disk: 500 GB Space

Software Requirements:

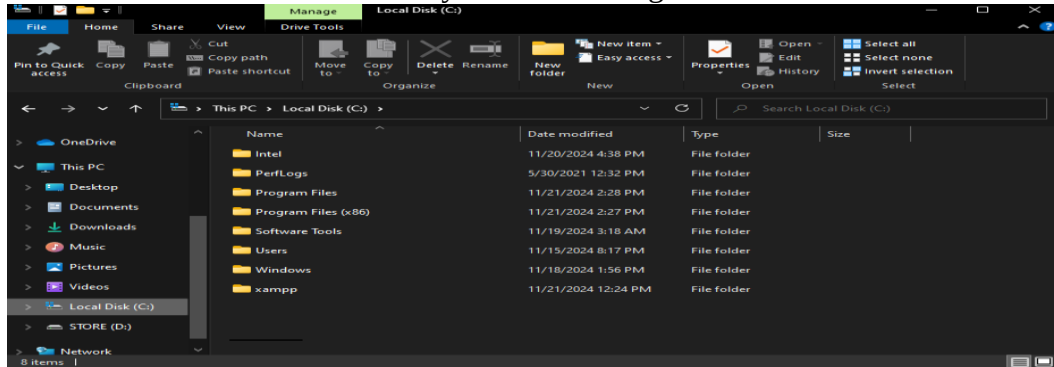
- ❖ Operating System: Windows /iOS/Unix 17
- ❖ Web Browser: Google Chrome/Firefox
- ❖ Technology: PHP
- ❖ Tools: XAMPP
- ❖ Web Design: HTML, CSS, JAVASCRIPT
- ❖ Back End: MYSQL
- ❖ Scripting Language: PHP
- ❖ Visual Studio Code or Notepad ++ for making minor change

Step-by-Step Instructions

To set up your online_car_rental-system project in XAMPP, follow these steps:

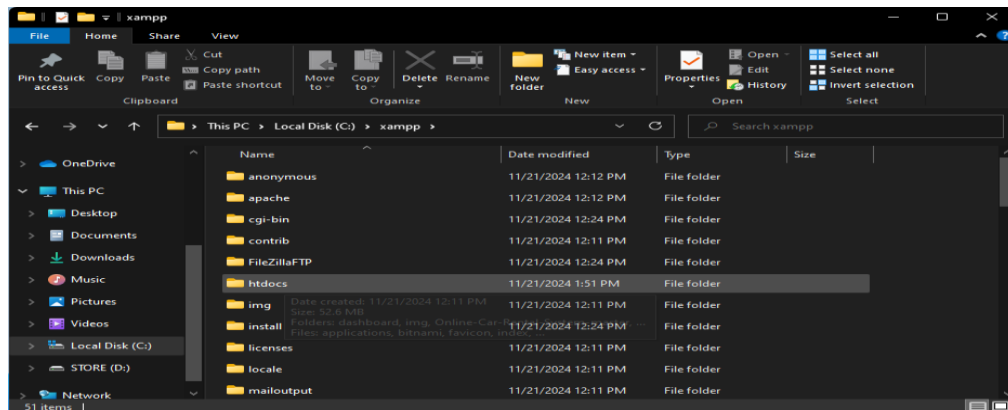
❖ Navigate to the XAMPP Folder

Open Local Disk C (C:), locate the xampp folder, and open it. This is where XAMPP keeps all its necessary files and configurations.



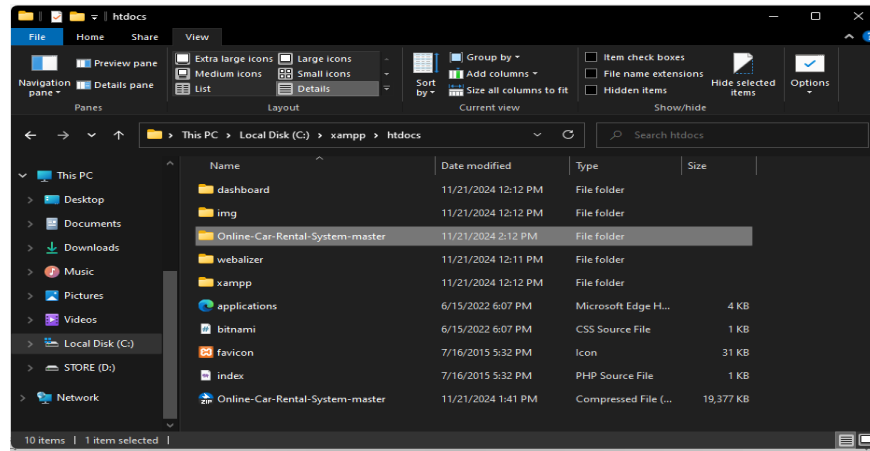
❖ Open the htdocs Folder

Inside the xampp folder, find the htdocs folder and open it. This folder serves as the root directory for all web projects.



❖ Copy Your Project

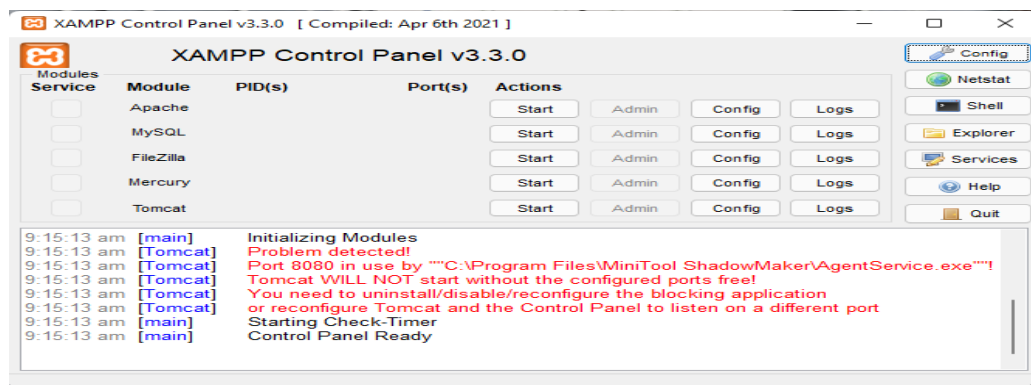
Copy the online_car_rental-system folder and paste it into the htdocs directory. The folder path should now look like C:\xampp\htdocs\online_car_rental-system.



1.2 HOW TO INSTALL THE APPLICATION

In order to run the system, a server such as Xampp server is needed. This can be downloaded from its official site <https://www.apachefriends.org/download.html> After it is done downloading, install the application.

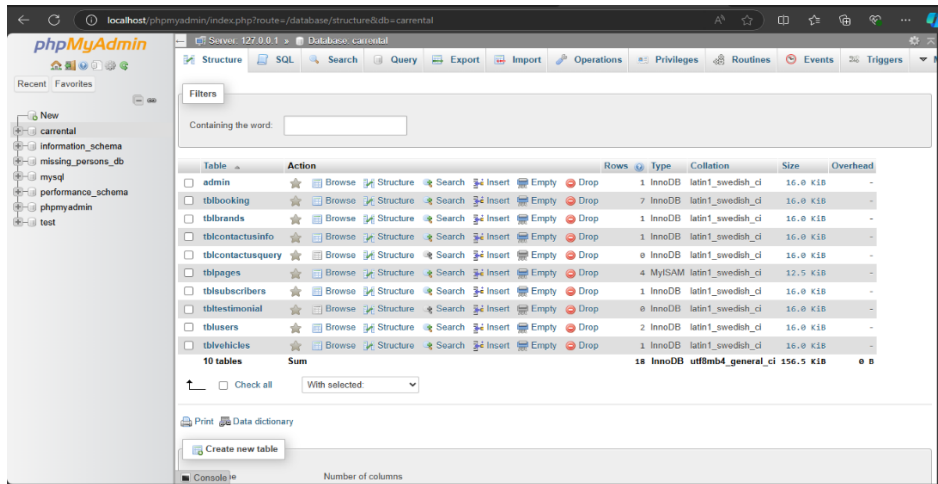
Once it is installed, launch Xampp either through the desktop icon or by searching for it using the windows search box. When it opens, it will look like the window below.



To activate the server, click start on the Apache and MSQL module. Once that is done, open up your browser and type in 127.0.0.1 which will take you to the dashboard as shown below.

Now copy the folder where the system is stored and paste in this location `C:\xampp\htdocs\`

After that is done, import the database of the system that will be found in a folder named Database that is within the system folder to Xampp server as shown below;



File Import

First create the name of the database and then import the SQL file. Once it is imported paste this URL [http://127.0.0.1/ Online_Car_Rental](http://127.0.0.1/Online_Car_Rental) into your browser and the system will execute.

Access Your Project

Open your web browser and type http://localhost/online_car_rental-system to run the project. If the project includes a database, ensure you set it up using phpMyAdmin and configure the database connection file accordingly

APPENDIX B

USER MANUAL

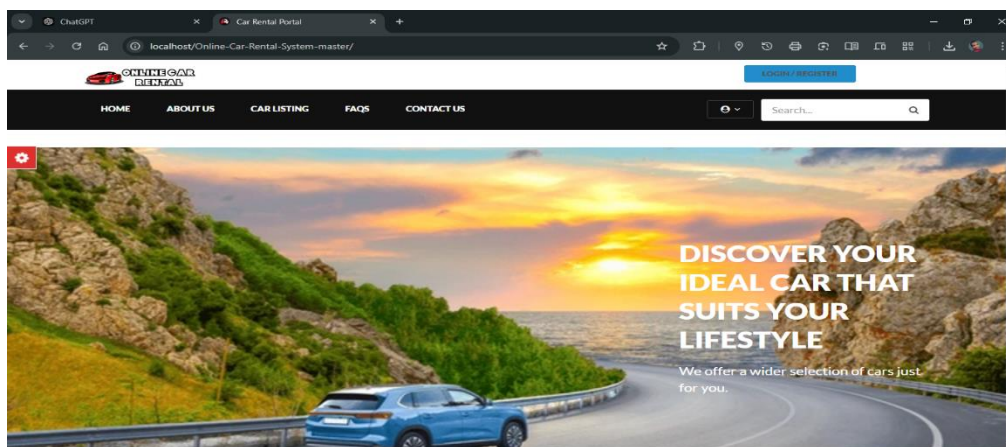
INTRODUCTION

A user manual is a document designed to assist users in navigating a specific system, product, or service seamlessly. Also known as an instruction manual or user guide, it provides detailed information on operations, standards, guidelines, functionalities, and troubleshooting.

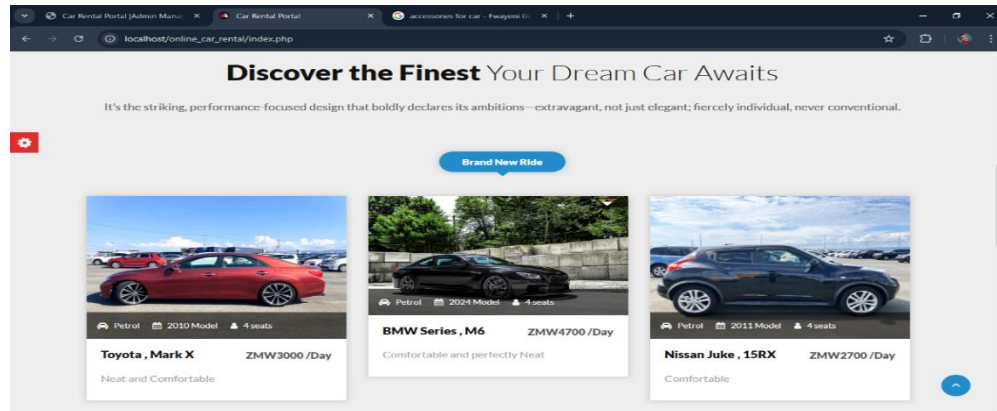
User manuals typically include step-by-step instructions that guide users on how to utilize the product effectively and address any issues that may arise. While it's not necessary for users to read the manual from cover to cover, it should include a table of contents and an index to help customers easily locate the relevant sections for their concerns.

Additionally, a "Getting Started" guide at the beginning of the manual allows users to quickly familiarize themselves with the product. User manuals can be delivered in printed form, online, or as a combination of both.

LANDING PAGE



1.1.1 LANDING PAGE



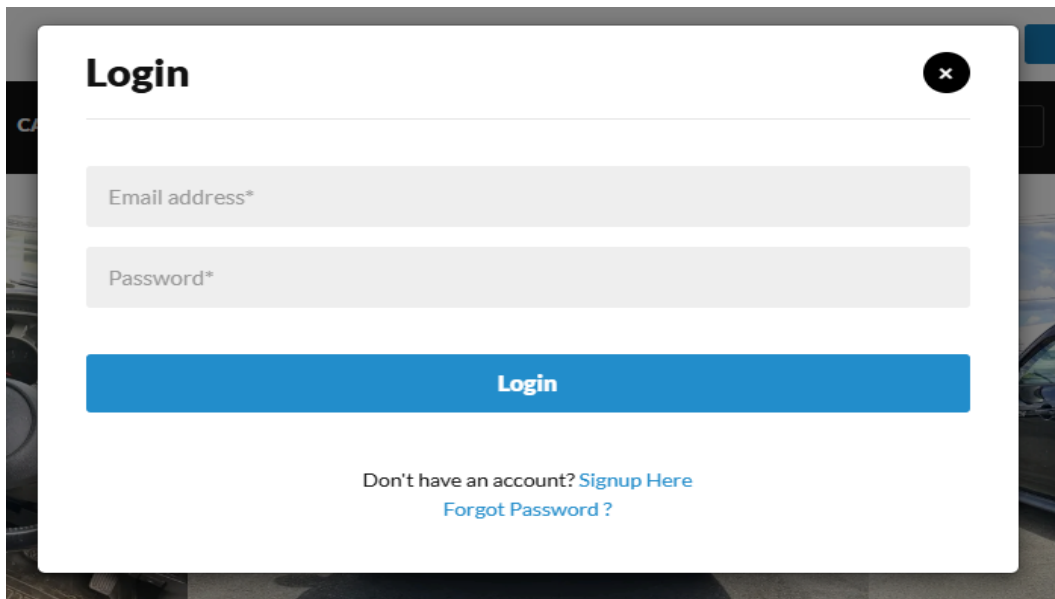
1.2 USER REGISTRATION

Before a user can rent a car, they must complete the registration process. This ensures that their details are stored securely in the system and can be referenced during future logins. The registration process is essential for creating a personalized experience and maintaining accurate records for all users.

A "Sign Up" form with a white background and a dark border. The form includes five input fields: "Full Name", "Mobile Number", "Email Address", "Password", and "Confirm Password". Below these fields is a checkbox labeled "I Agree with Terms and Conditions". A blue "Sign Up" button is at the bottom. A link "Already got an account? Login Here" is at the very bottom. A close button (X) is in the top right corner.

1.3 USER LOGIN

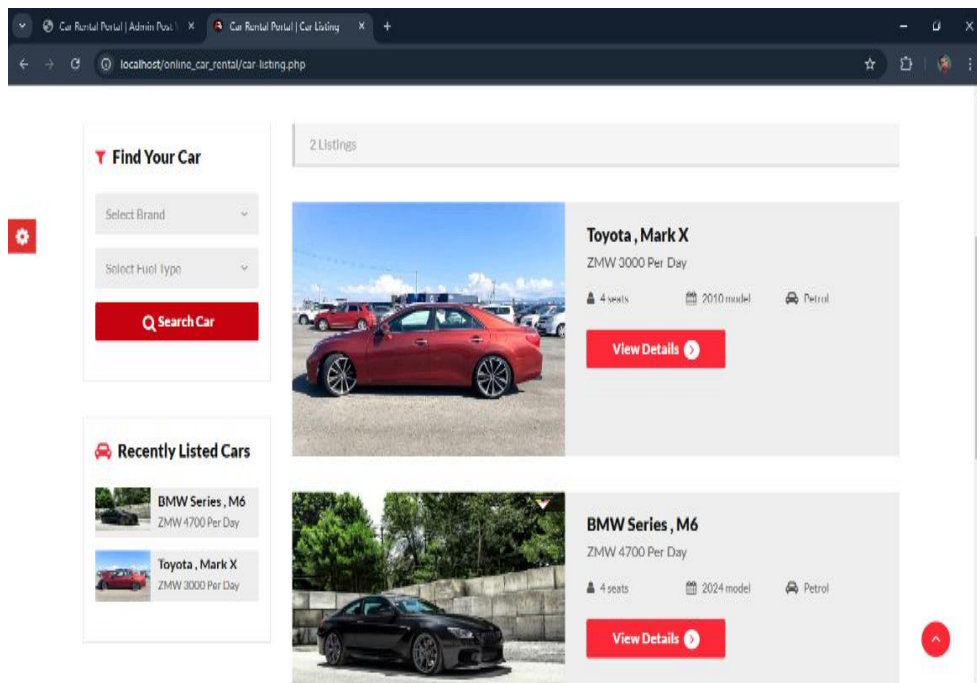
Users must log in before they can proceed to rent a car.



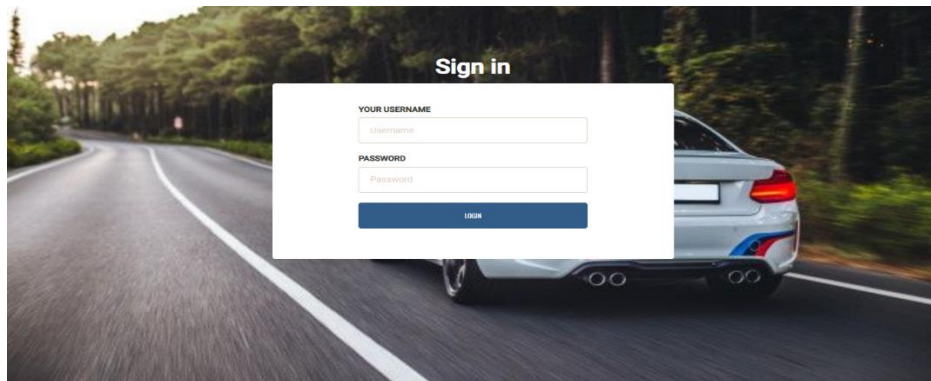
A login modal window with a white background and a dark border. It features a title bar with a close button (X) in the top right corner. The form contains two input fields: "Email address*" and "Password*", both with light gray borders. Below these fields is a prominent blue button labeled "Login". At the bottom of the modal, there is a link "Don't have an account? Signup Here" and another link "Forgot Password?" in blue text.

1.4 CAR LISTED

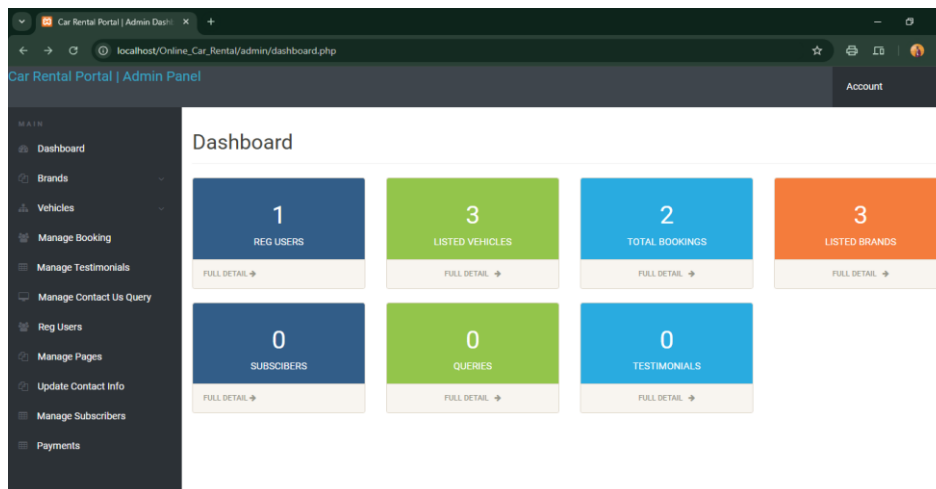
These are the vehicles that have been listed on the platform.



ADMIN LOGIN

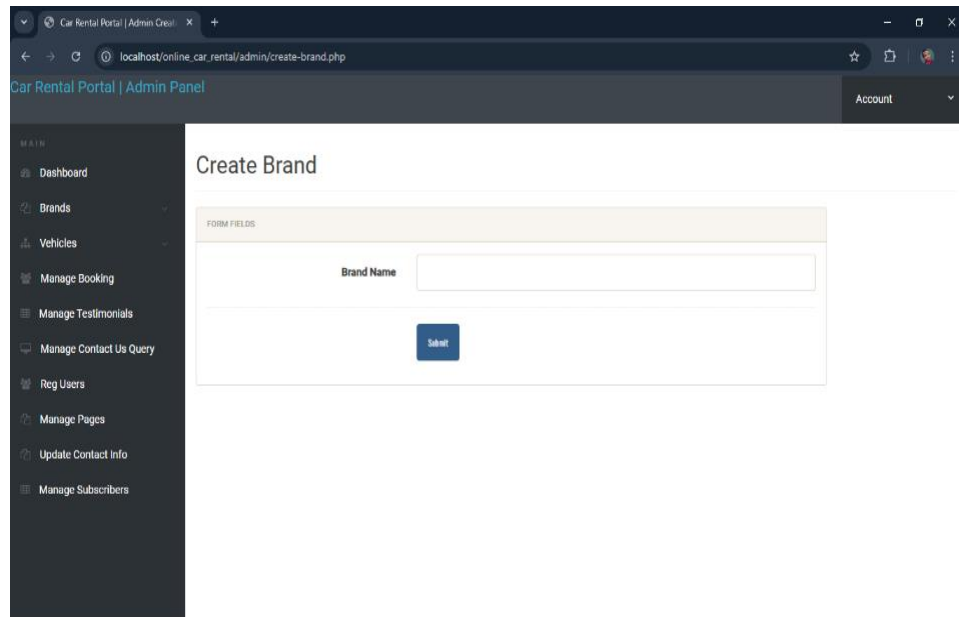


ADMIN DASHBOARD



ADD VEHICLE BRAND

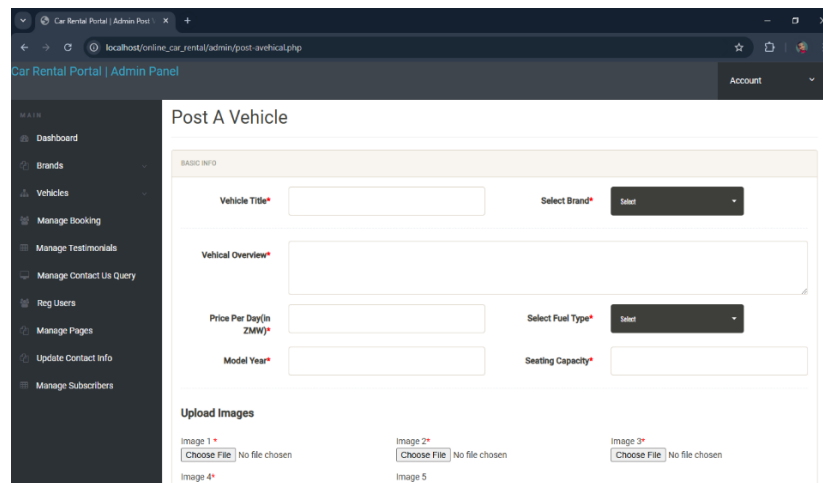
Using this webpage, the admin can add new vehicle brands to the system, expanding the range of options available for users to choose from.



The screenshot shows a web browser window with the URL `localhost/online_car_rental/admin/create-brand.php`. The page title is "Car Rental Portal | Admin Panel". On the left is a dark sidebar with a "MAIN" section containing links: Dashboard, Brands, Vehicles, Manage Booking, Manage Testimonials, Manage Contact Us Query, Reg Users, Manage Pages, Update Contact Info, and Manage Subscribers. The main content area is titled "Create Brand" and contains a "FORM FIELDS" section with a single text input labeled "Brand Name" and a blue "Submit" button below it.

POST VEHICLES

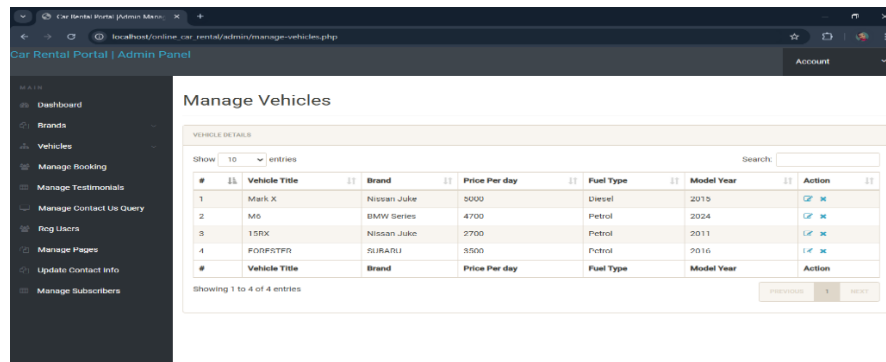
Using this webpage, the admin can add new vehicles to the system, keeping the vehicle inventory updated and expanding available options for users.



The screenshot shows a web browser window with the URL `localhost/online_car_rental/admin/post-avehical.php`. The page title is "Car Rental Portal | Admin Panel". On the left is a dark sidebar with a "MAIN" section containing links: Dashboard, Brands, Vehicles, Manage Booking, Manage Testimonials, Manage Contact Us Query, Reg Users, Manage Pages, Update Contact Info, and Manage Subscribers. The main content area is titled "Post A Vehicle" and contains a "BASIC INFO" section with the following fields: "Vehicle Title*" (text input), "Select Brand*" (dropdown menu with a "Select" button), "Vehical Overview*" (text area), "Price Per Day(in ZMW)*" (text input), "Select Fuel Type*" (dropdown menu with a "Select" button), "Model Year*" (text input), and "Seating Capacity*" (text input). Below these fields is an "Upload Images" section with five slots: "Image 1*", "Image 2*", "Image 3*", "Image 4*", and "Image 5". Each slot contains a "Choose File" button and the text "No file chosen".

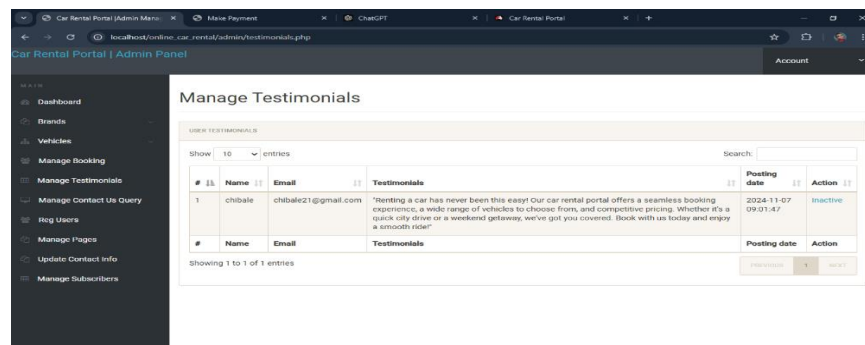
MANAGE VEHICLES

Using this webpage, the admin can edit or delete vehicles from the system, ensuring that vehicle listings remain accurate and up-to-date.



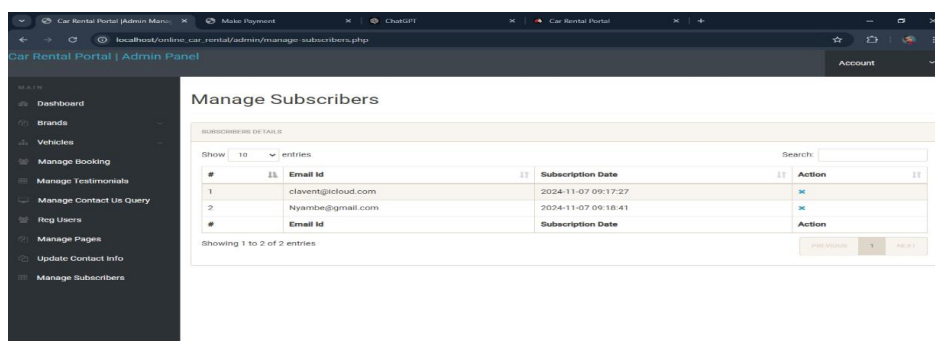
MANAGE TESTIMONIALS

Using this webpage, the admin reviews, approves, edits, or deletes testimonials to ensure only appropriate and relevant feedback is displayed on the website.



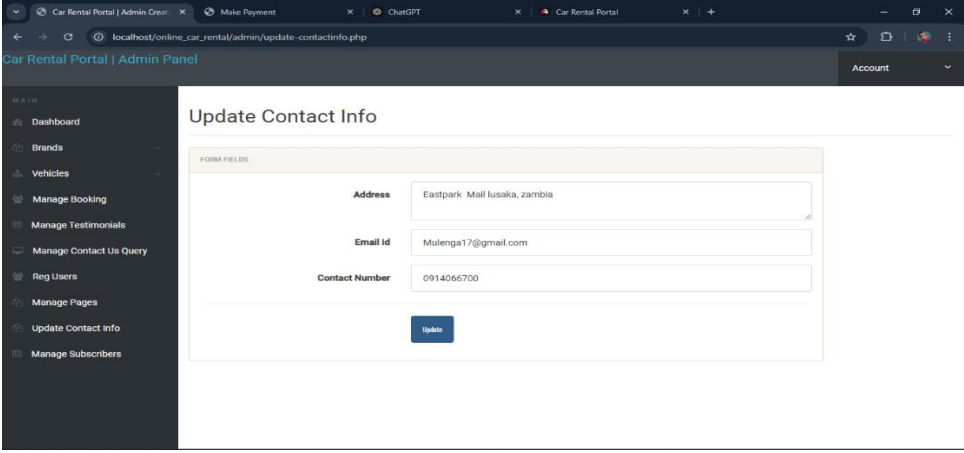
MANAGE SUBSCRIBERS

The admin manages subscriptions by adding, reviewing, and removing subscriber information to keep the mailing list accurate and up-to-date.



UPDATE CONTACT INFO

Using this webpage, the admin updates information by editing, adding, or removing content to ensure that all details are current and accurate on the platform.



Car Rental Portal | Admin Panel

Update Contact Info

FORM FIELDS

Address: Eastpark Mail lusaka, zambia

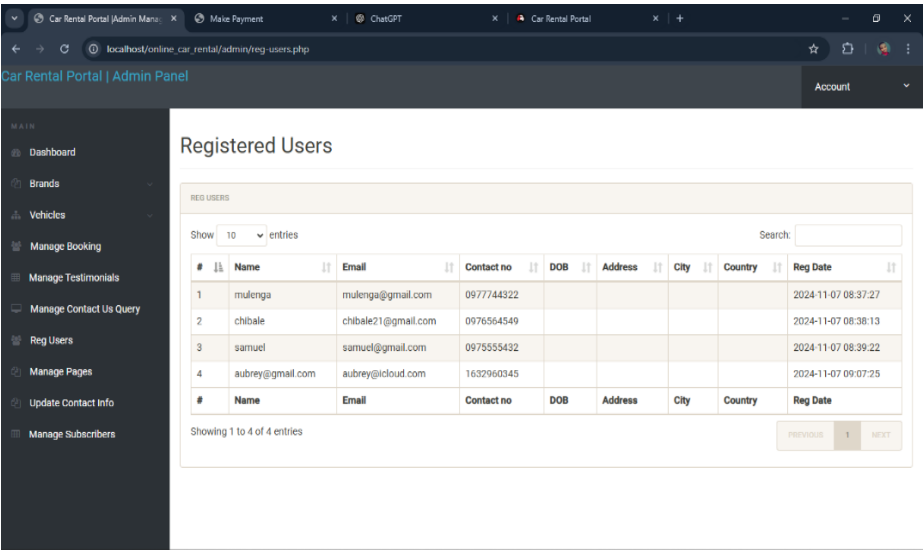
Email id: Mulenga17@gmail.com

Contact Number: 0914066700

Update

REGISTERED USER

The admin manages registered users by adding new users, reviewing user details, updating profiles, and removing inactive or unauthorized accounts to maintain a secure and active user base on the platform.



Car Rental Portal | Admin Panel

Registered Users

REG USERS

Show 10 entries

Search:

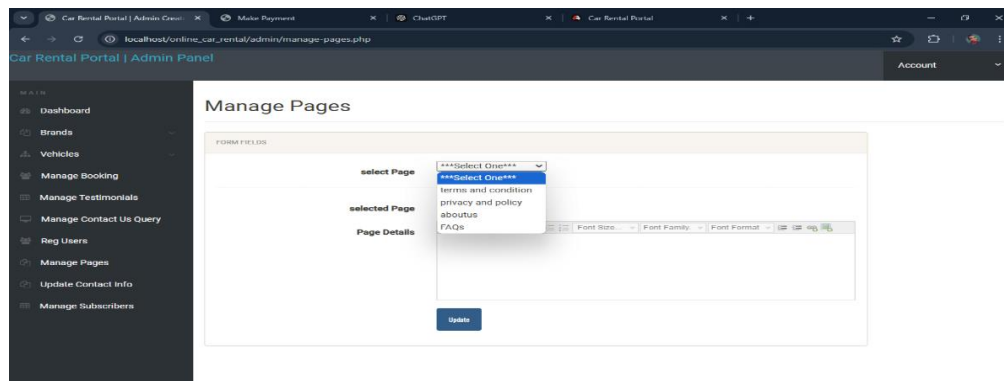
#	Name	Email	Contact no	DOB	Address	City	Country	Reg Date
1	mulenga	mulenga@gmail.com	0977744322					2024-11-07 08:37:27
2	chibale	chibale21@gmail.com	0976564549					2024-11-07 08:38:13
3	samuel	samuel@gmail.com	0975555432					2024-11-07 08:39:22
4	aubrey@gmail.com	aubrey@icloud.com	1632960345					2024-11-07 09:07:25

Showing 1 to 4 of 4 entries

PREVIOUS 1 NEXT

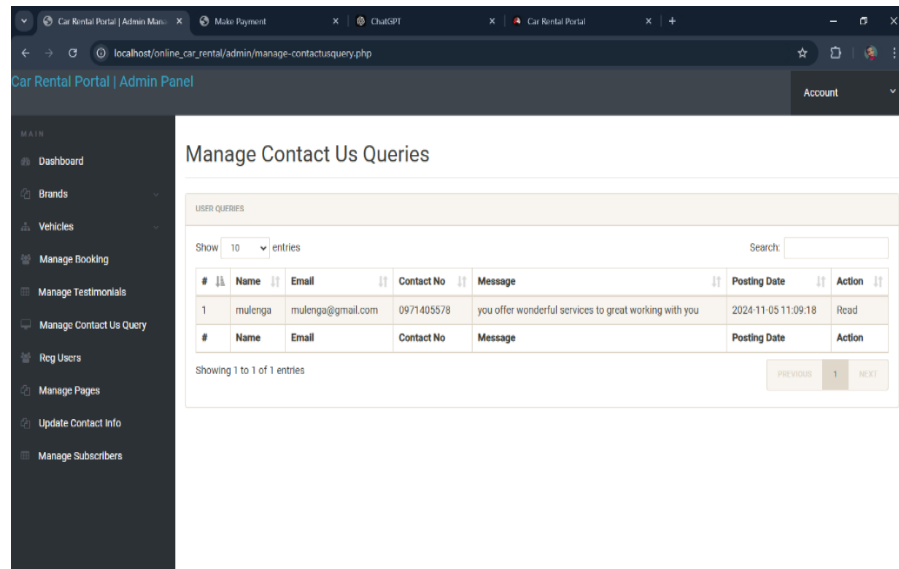
MANAGE PAGES

Using this webpage, the admin manages pages like Terms and Conditions, Privacy Policy, About Us, and FAQs by creating, updating, and editing content to ensure information is clear, accurate, and aligned with the platform's standards and policies



MANAGE CONTACT US QUERIES

Using this webpage, the admin manages Contact Us queries by reviewing messages from users, responding to inquiries, addressing concerns, and resolving issues to provide effective support and maintain user satisfaction.



PAYMENT

Car Rental Portal | Admin Panel

Account

MAIN

Dashboard

Brands

Vehicles

Manage Booking

Manage Testimonials

Manage Contact Us Query

Reg Users

Manage Pages

Update Contact Info

Manage Subscribers

Payments

Payments Report

Export to CSV

Manage Payments

PAYMENTS DETAILS

Show10entries

Search:

#	User Name	User Email	Vehicle Title	Amount	Payment Date	Booking Status	Action
1	SAMUEL	samuel@gmail.com	benz32	20044	2024-11-18	Confirmed	
#	User Name	User Email	Vehicle Title	Amount	Payment Date	Booking Status	Action

Showing 1 to 1 of 1 entries

PREVIOUS

1

NEXT

Report on Observations and Interviews for the Online Car Rental System

To inform the development of the Online Car Rental System, interviews and observations were conducted by Jimmy M Chiwele to gather insights into the current challenges and user expectations. The interviews were held with key stakeholders, including Mike Njovu from **ZUBA64**, Brian Mukuka from **Imuks App Solutions**, Danny, and Aubrey, from **YANGO** using a semi-structured format to explore pain points and preferences. Additionally, observational research was carried out at two local car rental agencies to understand customer behavior and the effectiveness of existing booking systems.

The observations revealed that the current booking processes at car rental agencies were inefficient, with long wait times and repetitive steps. Customers often expressed frustration with the outdated systems, especially with regard to mobile accessibility. They preferred a faster, more streamlined process and wanted clearer vehicle descriptions to aid in their selection. The existing systems were not optimized for mobile devices, which posed accessibility challenges for customers trying to book on the go.

From the interviews, several insights were gathered. Mike Njovu from *zuba64* suggested that the system include real-time feedback after booking and secure payment options to ensure user confidence. Brian Mukuka from *Imuks App Solutions* recommended that the system be easy to navigate with a well-structured layout and integrated with social media for wider outreach. Aubrey from *YANGO*, stressed the importance of making the system mobile-responsive and providing clear visual feedback during the booking process.

The combined findings highlighted the need for a user-friendly platform that is both efficient and secure. Key priorities include creating a streamlined booking process, ensuring mobile responsiveness, offering secure payment methods, and providing real-time updates to users throughout the booking process.

In conclusion, the research has provided valuable insights that will inform the design of the Online Car Rental System. The new platform will focus on improving user experience by addressing the current system's inefficiencies and ensuring that the needs and expectations of users are met with a simple, mobile-friendly, and secure solution.

APPENDIX C

TECHNICAL REPORT

SOURCE CODE

The following chapter lists functional code.

VEHICLE-DETAILS CODE

```
<?php
session_start();
include('includes/config.php');
error_reporting(0);

if(isset($_POST['submit']))
{
    $fromdate=$_POST['fromdate'];
    $todate=$_POST['todate'];
    $message=$_POST['message'];
    $useremail=$_SESSION['login'];
    $status=0;
    $vhid=$_GET['vhid'];
    $sql="INSERT INTO tblbooking(userEmail,VehicleId,FromDate,ToDate,message,Status)
VALUES(:useremail,:vhid,:fromdate,:todate,:message,:status)";
    $query = $dbh->prepare($sql);
    $query->bindParam(':useremail',$useremail,PDO::PARAM_STR);
    $query->bindParam(':vhid',$vhid,PDO::PARAM_STR);
    $query->bindParam(':fromdate',$fromdate,PDO::PARAM_STR);
    $query->bindParam(':todate',$todate,PDO::PARAM_STR);
    $query->bindParam(':message',$message,PDO::PARAM_STR);
    $query->bindParam(':status',$status,PDO::PARAM_STR);
    $query->execute();
    $lastInsertId = $dbh->lastInsertId();

    if(isset($_GET['vid'])){
        $_SESSION['vehicle_id'] = $_GET['vid'];
        try {
            // Prepare the database query to retrieve the user ID
            $sql = "SELECT id FROM tblusers WHERE EmailId=:email";
            $query = $dbh->prepare($sql);
            $query->bindParam(':email', $useremail);
            $query->execute();

            // Fetch the query results
```

```

$result = $query->fetch();

// Check if a user ID is found
if ($result) {
    // Store the user ID in the session
    $_SESSION['user_id'] = $result['id'];
}
} catch (PDOException $e) {
    // Handle database query errors
    echo 'Error fetching user ID: ' . $e->getMessage();
}
}
//redirect to a payment form
header('location:makepayments/index.php');
}

// echo "<script>console.log($lastInsertId);</script>";
// if($lastInsertId)
// {
// echo "<script>alert('Booking successfull.');

```

MANAGE PAGE CODE

```

<?php
session_start();
error_reporting(0);
include('includes/config.php');
if(strlen($_SESSION['alogin'])==0)
{
    header('location:index.php');
}
else{
    if($_POST['submit']=="Update")
    {
        $pagetype=$_GET['type'];
    }
}

```

```

    $pagedetails=$_POST['pgedetails'];
    $sql = "UPDATE tblpages SET detail=:pagedetails WHERE type=:pagetype";
    $query = $dbh->prepare($sql);
    $query -> bindParam(':pagetype',$pagetype, PDO::PARAM_STR);
    $query-> bindParam(':pagedetails',$pagedetails, PDO::PARAM_STR);
    $query -> execute();
    $msg="Page data updated successfully";

}

?>

```

MANAGE BOOKING CODE

```

<?php
session_start();
error_reporting(0);
include('includes/config.php');
if(strlen($_SESSION['alogin'])==0)
    {
header('location:index.php');
    }
else{
if(isset($_REQUEST['eid']))
    {
    $eid=intval($_GET['eid']);
    $status="2";
    $sql = "UPDATE tblbooking SET Status=:status WHERE id=:eid";
    $query = $dbh->prepare($sql);
    $query -> bindParam(':status',$status, PDO::PARAM_STR);
    $query-> bindParam(':eid',$eid, PDO::PARAM_STR);
    $query -> execute();

    $msg="Booking Successfully Cancelled";
    }

if(isset($_REQUEST['aaid']))
    {
    $aaid=intval($_GET['aaid']);
    $status=1;

    $sql = "UPDATE tblbooking SET Status=:status WHERE id=:aaid";
    $query = $dbh->prepare($sql);
    $query -> bindParam(':status',$status, PDO::PARAM_STR);
    $query-> bindParam(':aaid',$aaid, PDO::PARAM_STR);
    $query -> execute();

```

```
$msg="Booking Successfully Confirmed";  
}
```

```
?>
```

CREATE BRAND CODE

```
<?php  
session_start();  
error_reporting(0);  
include('includes/config.php');  
if(strlen($_SESSION['alogin'])==0)  
{  
header('location:index.php');  
}  
else{  
// Code for change password  
if(isset($_POST['submit']))  
{  
$brand=$_POST['brand'];  
$sql="INSERT INTO tblbrands(BrandName) VALUES(:brand)";  
$query = $dbh->prepare($sql);  
$query->bindParam(':brand',$brand,PDO::PARAM_STR);  
$query->execute();  
$lastInsertId = $dbh->lastInsertId();  
if($lastInsertId)  
{  
$msg="Brand Created successfully";  
}  
else  
{  
$error="Something went wrong. Please try again";  
}  
}  
?>
```

MANAGE CONTACT US QUERY

```
<?php
session_start();
error_reporting(0);
include('includes/config.php');

if(strlen($_SESSION['alogin'])==0) {
    header('location:index.php');
} else {
    if(isset($_REQUEST['eid'])) {
        $eid = intval($_GET['eid']);
        $status = 1; // Mark as read
        $sql = "UPDATE tblcontactusquery SET status=:status WHERE id=:eid";
        $query = $dbh->prepare($sql);
        $query->bindParam(':status', $status, PDO::PARAM_STR);
        $query->bindParam(':eid', $eid, PDO::PARAM_STR);
        $query->execute();

        $msg = "Query successfully marked as read";
    }
?>
```

MANAGE SUBSCRIBERS

```
<?php
session_start();
error_reporting(0);
include('includes/config.php');
if(strlen($_SESSION['alogin'])==0)
{
    header('location:index.php');
}
else{
    if(isset($_GET['del']))
    {
        $id=$_GET['del'];
        $sql = "delete from tblsubscribers WHERE id=:id";
        $query = $dbh->prepare($sql);
        $query -> bindParam(':id',$id, PDO::PARAM_STR);
        $query -> execute();
        $msg="Subscriber info deleted";
    }
?>
```

POST VEHICLES CODE

```
<?php
session_start();
error_reporting(0);
include('includes/config.php');
if(strlen($_SESSION['alogin'])==0)
    {
header('location:index.php');
    }
else{

if(isset($_POST['submit']))
    {
$vehicletitle=$_POST['vehicletitle'];
$brand=$_POST['brandname'];
$vehicleoverview=$_POST['vehicalorcview'];
$priceperday=$_POST['priceperday'];
$fueltype=$_POST['fueltype'];
$modelyear=$_POST['modelyear'];
$seatingcapacity=$_POST['seatingcapacity'];
$vimage1=$_FILES["img1"]["name"];
$vimage2=$_FILES["img2"]["name"];
$vimage3=$_FILES["img3"]["name"];
$vimage4=$_FILES["img4"]["name"];
$vimage5=$_FILES["img5"]["name"];
$airconditioner=$_POST['airconditioner'];
$powerdoorlocks=$_POST['powerdoorlocks'];
$antilockbrakingsys=$_POST['antilockbrakingsys'];
$brakeassist=$_POST['brakeassist'];
$powersteering=$_POST['powersteering'];
$driverairbag=$_POST['driverairbag'];
$passengerairbag=$_POST['passengerairbag'];
$powerwindow=$_POST['powerwindow'];
$cdplayer=$_POST['cdplayer'];
$centrallocking=$_POST['centrallocking'];
$crashcensor=$_POST['crashcensor'];
$leatherseats=$_POST['leatherseats'];
move_uploaded_file($_FILES["img1"]["tmp_name"],"img/vehicleimages/".$_FILES["img1"]["name"]);
move_uploaded_file($_FILES["img2"]["tmp_name"],"img/vehicleimages/".$_FILES["img2"]["name"]);
move_uploaded_file($_FILES["img3"]["tmp_name"],"img/vehicleimages/".$_FILES["img3"]["name"]);
move_uploaded_file($_FILES["img4"]["tmp_name"],"img/vehicleimages/".$_FILES["img4"]["name"]);
    }
```

```
move_uploaded_file($_FILES["img5"]["tmp_name"],"img/vehicleimages/".$_FILES["img5"]["name"]);
```

```
$sql="INSERT
tblvehicles(VehiclesTitle,VehiclesBrand,VehiclesOverview,PricePerDay,FuelType,ModelYear,SeatingCapacity,Vimage1,Vimage2,Vimage3,Vimage4,Vimage5,AirConditioner,PowerDoorLocks,AntiLockBrakingSystem,BrakeAssist,PowerSteering,DriverAirbag,PassengerAirbag,PowerWindows,CDPlayer,CentralLocking,CrashSensor,LeatherSeats)
VALUES(:vehicletitle,:brand,:vehicleoverview,:priceperday,:fueltype,:modelyear,:seatingcapacity,:vimage1,:vimage2,:vimage3,:vimage4,:vimage5,:airconditioner,:powerdoorlocks,:antilockbrakingsys,:brakeassist,:powersteering,:driverairbag,:passengerairbag,:powerwindow,:cdplayer,:centrallocking,:crashcensor,:leatherseats)";
$query = $dbh->prepare($sql);
$query->bindParam(':vehicletitle',$vehicletitle,PDO::PARAM_STR);
$query->bindParam(':brand',$brand,PDO::PARAM_STR);
$query->bindParam(':vehicleoverview',$vehicleoverview,PDO::PARAM_STR);
$query->bindParam(':priceperday',$priceperday,PDO::PARAM_STR);
$query->bindParam(':fueltype',$fueltype,PDO::PARAM_STR);
$query->bindParam(':modelyear',$modelyear,PDO::PARAM_STR);
$query->bindParam(':seatingcapacity',$seatingcapacity,PDO::PARAM_STR);
$query->bindParam(':vimage1',$vimage1,PDO::PARAM_STR);
$query->bindParam(':vimage2',$vimage2,PDO::PARAM_STR);
$query->bindParam(':vimage3',$vimage3,PDO::PARAM_STR);
$query->bindParam(':vimage4',$vimage4,PDO::PARAM_STR);
$query->bindParam(':vimage5',$vimage5,PDO::PARAM_STR);
$query->bindParam(':airconditioner',$airconditioner,PDO::PARAM_STR);
$query->bindParam(':powerdoorlocks',$powerdoorlocks,PDO::PARAM_STR);
$query->bindParam(':antilockbrakingsys',$antilockbrakingsys,PDO::PARAM_STR);
$query->bindParam(':brakeassist',$brakeassist,PDO::PARAM_STR);
$query->bindParam(':powersteering',$powersteering,PDO::PARAM_STR);
$query->bindParam(':driverairbag',$driverairbag,PDO::PARAM_STR);
$query->bindParam(':passengerairbag',$passengerairbag,PDO::PARAM_STR);
$query->bindParam(':powerwindow',$powerwindow,PDO::PARAM_STR);
$query->bindParam(':cdplayer',$cdplayer,PDO::PARAM_STR);
$query->bindParam(':centrallocking',$centrallocking,PDO::PARAM_STR);
$query->bindParam(':crashcensor',$crashcensor,PDO::PARAM_STR);
$query->bindParam(':leatherseats',$leatherseats,PDO::PARAM_STR);
$query->execute();
$lastInsertId = $dbh->lastInsertId();
if($lastInsertId)
{
$msg="Vehicle posted successfully";
}
else
{
$error="Something went wrong. Please try again";
```

```
}
```

```
}
```

```
?>
```

REGISTERED USER CODE

```
<?php
session_start();
error_reporting(0);
include('includes/config.php');
if(strlen($_SESSION['alogin'])==0)
{
header('location:index.php');
}
else{
if(isset($_GET['del']))
{
$id=$_GET['del'];
$sql = "delete from tblbrands WHERE id=:id";
$query = $dbh->prepare($sql);
$query -> bindParam(':id',$id, PDO::PARAM_STR);
$query -> execute();
$msg="Page data updated successfully";

}


```

```
?>
```

PAYMENT CODE

```
<?php
if(isset($_POST['pay']))
{
    $email = $_POST['email'];
    $amount = $_POST['amount'];
    /* Prepare our rave request
    $request = [
        'tx_ref' => time(),
        'amount' => $amount,
        'currency' => 'ZMW',
        'payment_options' => 'mobilemoneyzambia',
        'redirect_url' => 'http://localhost/Online_Car_Rental/makepayments/process.php',

```

```

        'customer' => [
            'email' => $email,
            'name' => ''
        ],
        'meta' => [
            'price' => $amount
        ],
        'customizations' => [
            'title' => 'Paying for a sample product',
            'description' => 'sample'
        ]
    ];
//flutterwave endpoint
    $curl = curl_init();

    curl_setopt_array($curl, array(
        CURLOPT_URL => 'https://api.flutterwave.com/v3/payments',
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_ENCODING => "",
        CURLOPT_MAXREDIRS => 10,
        CURLOPT_TIMEOUT => 0,
        CURLOPT_FOLLOWLOCATION => true,
        CURLOPT_HTTP_VERSION => CURL_HTTP_VERSION_1_1,
        CURLOPT_CUSTOMREQUEST => 'POST',
        CURLOPT_POSTFIELDS => json_encode($request),
        CURLOPT_HTTPHEADER => array(
            'Content-Type: application/json'
        ),
    ));
    $response = curl_exec($curl);

    curl_close($curl);

    $res = json_decode($response);
    if($res->status == 'success')
    {
        $link = $res->data->link;
        header('Location: '.$link);
    }
    else
    {
        echo '<script>alert("We can not process your payment");</script>';
    }
}
?>

```

CHECK AVAILABILITY CODE

```
<?php
require_once("includes/config.php");
// code user email availability
if(!empty($_POST["emailid"])) {
    $email= $_POST["emailid"];
    if (filter_var($email, FILTER_VALIDATE_EMAIL)===false) {

        echo "error : You did not enter a valid email.";
    }
    else {
        $sql ="SELECT EmailId FROM tblusers WHERE EmailId=:email";
        $query= $dbh -> prepare($sql);
        $query-> bindParam(':email', $email, PDO::PARAM_STR);
        $query-> execute();
        $results = $query -> fetchAll(PDO::FETCH_OBJ);
        $cnt=1;
        if($query -> rowCount() > 0)
        {
            echo "<span style='color:red'> Email already exists .</span>";
            echo "<script>$('#submit').prop('disabled',true);</script>";
        } else{

            echo "<span style='color:green'> Email available for Registration .</span>";
            echo "<script>$('#submit').prop('disabled',false);</script>";
        }
    }
}

?>
```

EDIT VEHICLE CODE

```
<?php
session_start();
error_reporting(E_ALL); // Enable full error reporting to catch issues
ini_set('display_errors', 1);

include('includes/config.php');
if(strlen($_SESSION['alogin'])==0) {
    header('location:index.php');
} else {

    if(isset($_POST['submit'])) {
        // Get values from form input
```

```

$vehicletitle = $_POST['vehicletitle'];
$brand = $_POST['brandname'];
$vehicleoverview = $_POST['vehicalorcview'];
$priceperday = $_POST['priceperday'];
$fueltype = $_POST['fueltype'];
$modelyear = $_POST['modelyear'];
$seatingcapacity = $_POST['seatingcapacity'];
$airconditioner = isset($_POST['airconditioner']) ? 1 : 0;
$powerdoorlocks = isset($_POST['powerdoorlocks']) ? 1 : 0;
$antilockbrakingsys = isset($_POST['antilockbrakingsys']) ? 1 : 0;
$brakeassist = isset($_POST['brakeassist']) ? 1 : 0;
$powersteering = isset($_POST['powersteering']) ? 1 : 0;
$driverairbag = isset($_POST['driverairbag']) ? 1 : 0;
$passengerairbag = isset($_POST['passengerairbag']) ? 1 : 0;
$powerwindow = isset($_POST['powerwindow']) ? 1 : 0;
$cdplayer = isset($_POST['cdplayer']) ? 1 : 0;
$centrallocking = isset($_POST['centrallocking']) ? 1 : 0;
$crashcensor = isset($_POST['crashcensor']) ? 1 : 0;
$leatherseats = isset($_POST['leatherseats']) ? 1 : 0;
$id = intval($_GET['id']);

```

```
// Update query
```

```

$sql = "UPDATE tblvehicles SET
    VehiclesTitle = :vehicletitle,
    VehiclesBrand = :brand,
    VehiclesOverview = :vehicleoverview,
    PricePerDay = :priceperday,
    FuelType = :fueltype,
    ModelYear = :modelyear,
    SeatingCapacity = :seatingcapacity,
    AirConditioner = :airconditioner,
    PowerDoorLocks = :powerdoorlocks,
    AntiLockBrakingSystem = :antilockbrakingsys,
    BrakeAssist = :brakeassist,
    PowerSteering = :powersteering,
    DriverAirbag = :driverairbag,
    PassengerAirbag = :passengerairbag,
    PowerWindows = :powerwindow,
    CDPlayer = :cdplayer,
    CentralLocking = :centrallocking,
    CrashSensor = :crashcensor,
    LeatherSeats = :leatherseats
    WHERE id = :id";
$query = $dbh->prepare($sql);

```

```
// Bind parameters
```

```

$query->bindParam(':vehicletitle', $vehicletitle, PDO::PARAM_STR);
$query->bindParam(':brand', $brand, PDO::PARAM_STR);
$query->bindParam(':vehicleoverview', $vehicleoverview, PDO::PARAM_STR);
$query->bindParam(':priceperday', $priceperday, PDO::PARAM_STR);
$query->bindParam(':fueltype', $fueltype, PDO::PARAM_STR);
$query->bindParam(':modelyear', $modelyear, PDO::PARAM_STR);
$query->bindParam(':seatingcapacity', $seatingcapacity, PDO::PARAM_STR);
$query->bindParam(':airconditioner', $airconditioner, PDO::PARAM_INT);
$query->bindParam(':powerdoorlocks', $powerdoorlocks, PDO::PARAM_INT);
$query->bindParam(':antilockbrakingsys', $antilockbrakingsys, PDO::PARAM_INT);
$query->bindParam(':brakeassist', $brakeassist, PDO::PARAM_INT);
$query->bindParam(':powersteering', $powersteering, PDO::PARAM_INT);
$query->bindParam(':driverairbag', $driverairbag, PDO::PARAM_INT);
$query->bindParam(':passengerairbag', $passengerairbag, PDO::PARAM_INT);
$query->bindParam(':powerwindow', $powerwindow, PDO::PARAM_INT);
$query->bindParam(':centrallocking', $centrallocking, PDO::PARAM_INT);
$query->bindParam(':crashcensor', $crashcensor, PDO::PARAM_INT);
$query->bindParam(':leatherseats', $leatherseats, PDO::PARAM_INT);
$query->bindParam(':id', $id, PDO::PARAM_INT);
// Execute and check for success
if($query->execute()) {
    $msg = "Data updated successfully";
} else {
    $msg = "Update failed";
}
}
}
?>

```

UPDATE PASSWORD CODE

```

<?php
session_start();
error_reporting(0);
include('includes/config.php');
if(strlen($_SESSION['login'])==0)
{
header('location:index.php');
}
else{
if(isset($_POST['update']))
{
$password=md5($_POST['password']);
$newpassword=md5($_POST['newpassword']);
$email=$_SESSION['login'];
$sql ="SELECT Password FROM tblusers WHERE EmailId=:email and Password=:password";

```

```

$query= $dbh -> prepare($sql);
$query-> bindParam(':email', $email, PDO::PARAM_STR);
$query-> bindParam(':password', $password, PDO::PARAM_STR);
$query-> execute();
$results = $query -> fetchAll(PDO::FETCH_OBJ);
if($query -> rowCount() > 0)
{
$con="update tblusers set Password=:newpassword where EmailId=:email";
$chngpwd1 = $dbh->prepare($con);
$chngpwd1-> bindParam(':email', $email, PDO::PARAM_STR);
$chngpwd1-> bindParam(':newpassword', $newpassword, PDO::PARAM_STR);
$chngpwd1->execute();
$msgg="Your Password succesfully changed";
}
else {
$error="Your current password is wrong";
}
}
?

```

LOGIN PAGE CODE

```

<?php
if(isset($_POST['login']))
{
$email=$_POST['email'];
$password=md5($_POST['password']);
$sql ="SELECT EmailId,Password,FullName FROM tblusers WHERE EmailId=:email and
Password=:password";
$query= $dbh -> prepare($sql);
$query-> bindParam(':email', $email, PDO::PARAM_STR);
$query-> bindParam(':password', $password, PDO::PARAM_STR);
$query-> execute();
$results=$query->fetchAll(PDO::FETCH_OBJ);
if($query->rowCount() > 0)
{
$_SESSION['login']=$_POST['email'];
$_SESSION['fname']=$results->FullName;
$currentpage=$_SERVER['REQUEST_URI'];
echo "<script type='text/javascript'> document.location = '$currentpage'; </script>";
} else{
echo "<script>alert('Invalid Details');</script>";
}
}

?>

```

